TOYOTA STARLET EP82

Wiring connections, Connector IDs & Loom Layouts

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This document details a large amount of wiring information which I could ascertain from the two main wiring looms (Cabin & Engine Bay) that was provided with a Right-Hand Drive 1991 (Gen 1) EP82 Toyota Starlet GT (4E-FTE) front cut, however much of the wiring layout and information will be appropriate to later Toyota Starlet models (with e.g. slightly different connectors or wiring colours).

Please note that there are no specific details of wiring connections and connectors within the doors, nor rear wiring loom of the car - due to those parts of the car not included with the front cut. The front cut was a Manual Transmission 4E-FTE which was not fitted with TEMS (Toyota Electronic Modulated Suspension) or ABS, so limited details of those components could be provided for those components.

To advise of any corrections, please contact me on the above email address.

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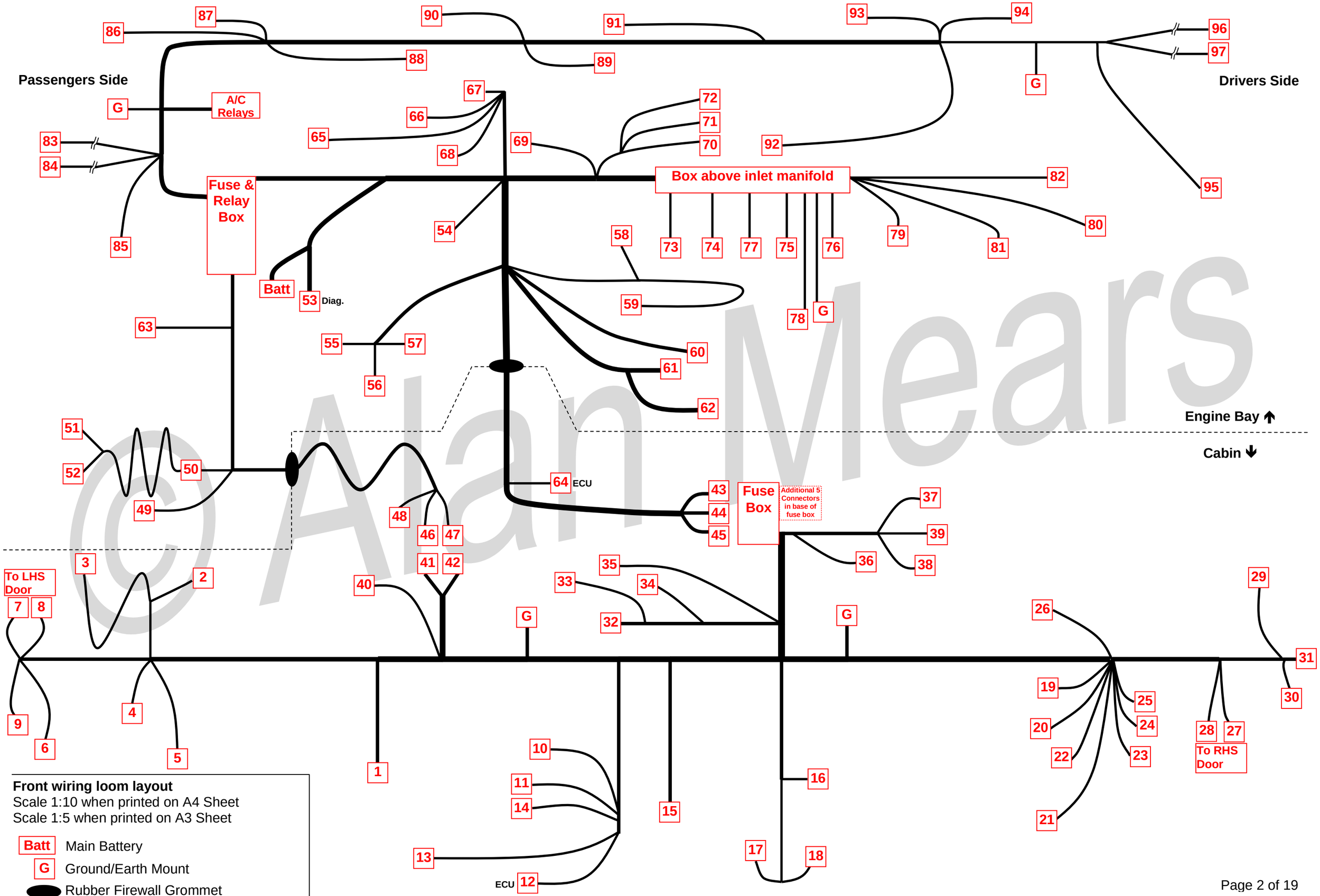

Front wiring loom layout
Scale 1:10 when printed on A4 Sheet
Scale 1:5 when printed on A3 Sheet

Batt	Main Battery
G	Ground/Earth Mount
●	Rubber Firewall Grommet

Connector ID Quick Reference Table

#	Cabin Loom	#	Engine Bay Loom	#	Engine Bay Loom
1	Circuit Opening Relay	**43**	To Cabin Fuse Box	**85**	A/C Pressure Switch
2	Fan/Blower	**44**	To Cabin Fuse Box	**86**	Front-Left Main Light
3	Fan/Heater Assembly	**45**	To Cabin Loom	**87**	Front-Left Park Light
4	TEMS Computer / Module	**46**	To Cabin Loom (Connector 41)	**88**	Radiator Fan
5	LHD Speaker (Cabin)	**47**	To Cabin Loom (Connector 42)	**89**	A/C Condenser Fan
6	Interior Light Door Switch (Left Door)	**48**	HVAC Module	**90**	Ambient Temperature Sensor
7	To Power Windows (Left Door)	**49**	Windscreen Wiper Motor (Front)	**91**	Horn
8	To Power Mirrors (Left Door)	**50**	Sub-harness Connector	**92**	A/C Compressor
9	Roof / Dome / Interior Light	**51**	Vacuum Switch Valve (HVAC)	**93**	Front-Right Main Light
10	Heater & A/C Dash Controls (Fan, Vent Direction, Recirc/Fresh Air)	**52**	Idle up Valve (HVAC)	**94**	Front-Right Park Light
11	Air vent Direction Motor	**53**	Diagnostics Port (Hardwired)	**95**	Windscreen Washer Pump
12	ECU	**54**	Reverse Switch on Gearbox	**96**	Front-Right Indicator
13	Cigarette Lighter	**55**	Ignitor	**97**	Front-Right Fog Light
14	Stereo Rear plug	**56**	Coil		
15	Clock (Upper Dash)	**57**	Unknown		
16	Instrument Cluster (Blue Connector)	**58**	Throttle Position Sensor		
17	Instrument Cluster (White Connector)	**59**	Boost Pressure / MAP Sensor		
18	Instrument Cluster (Boost Pressure Gauge & Lo/Hi Light)	**60**	Starter Motor Solenoid (Main Voltage)		
19	Turn Signal / Indicator Flasher Module	**61**	Starter Motor Solenoid (Trigger)		
20	Relay Light Reminder Module	**62**	Alternator		
21	Brake Pedal Switch	**63**	TEMS connector (LHS)		
22	Door Control Module	**64**	ECU		
23	RHS Speaker (Cabin)	**65**	Oil Pressure Sender		
24	Fog Light On/Off Switch	**66**	Catalytic Converter Overheat Sensor		
25	TEMS Switch	**67**	Oxygen / O2 Sensor		
26	Relay (hardwired) Power Windows	**68**	Coolant Sensor (To HVAC module)		
27	To Power Windows (Right Door)	**69**	Coolant Switch (For Radiator Thermofan)		
28	To Power Mirrors (Right Door)	**70**	Coolant Temperature Sensor (For Dash Temp Gauge)		
29	Interior Light Door Switch (Right Door)	**71**	Coolant Temperature Sensor (To ECU)		
30	To Rear Wiring Loom	**72**	Distributor		
31	Power Window Control Buttons	**73**	Injector 4		
32	Hazard Light Button	**74**	Injector 3		
33	Lo/Hi Boost Button	**75**	Injector 2		
34	Rear Window Demister Button	**76**	Injector 1		
35	Power Mirrors Adjustment Buttons	**77**	Inlet Manifold Temperature Sensor		
36	Ignition Barrel	**78**	Idle Up Valve		
37	Steering column combination switch (Rear Wiper/Washer)	**79**	Vacuum Solenoid Valve		
38	Steering column combination switch (Front Wipers/Washer)	**80**	Knock Sensor		
39	Steering column combination switch (Lights & Indicators)	**81**	Low Level Brake Fluid Reservoir Switch		
40	Foot/Lower Air Direction Motor	**82**	TEMS connector (RHS)		
41	To Engine Bay Loom (Connector 46)	**83**	Front-Left Indicator		
42	To Engine Bay Loom (Connector 47)	**84**	Front-Left Fog Light		

Connectors are referenced in the following pages with a '*C-*' *prefix* eg *C-12 for Connector 12*

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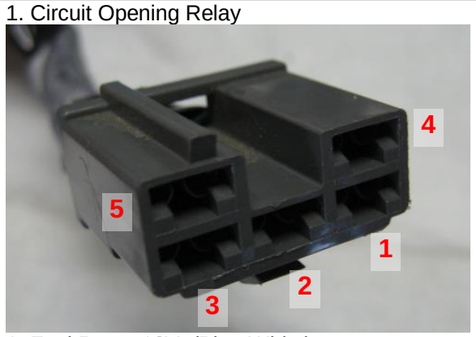
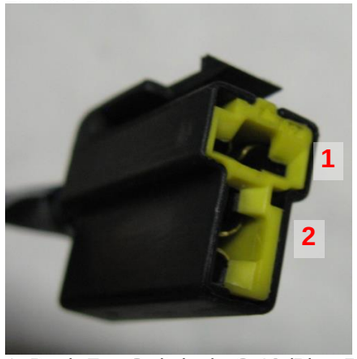
1. Circuit Opening Relay



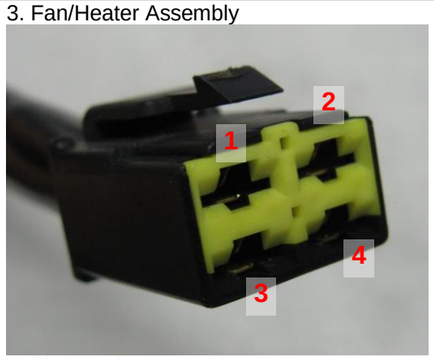
1. Fuel Pump 12V (Blue-White)
2. 12V via EFI Relay & ECU 'BATT' (Black-Red)
3. ECU 'STA'; Connector 36 'ST1' & Starter Motor Solenoid (Black)
4. ECU 'FC' (Green)
5. GND (White-Black)

2. Fan / Blower



1. Dash Fan Switch via *C-10* (Blue-Red)
2. From 'Heater" Fuse (Blue-White)

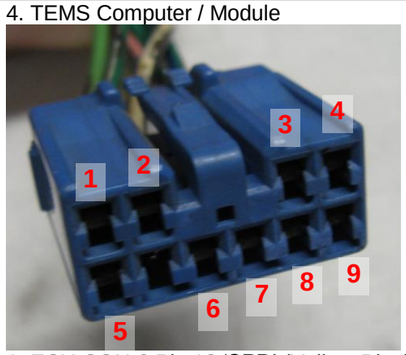
3. Fan/Heater Assembly



1. Blue-Red
2. Blue-Yellow
3. Blue-Black
4. White-Red

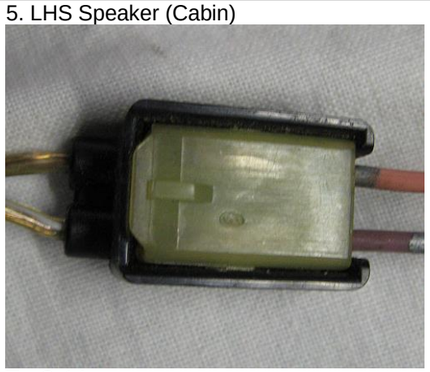
All wires from *C-10* (Dash Buttons)

4. TEMS Computer / Module



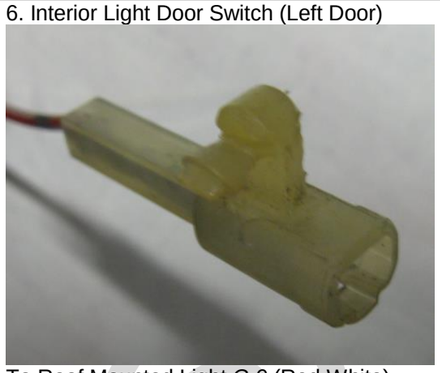
1. ECU CON-2 Pin 13 'SPD' (Yellow-Blue)
2. Green-White to *C-21*
3. Green-Yellow to *C-30, C-63* & *C-82*
4. From 'Ignition' Fuse (Blue-Green) to *C-30*
5. Light Green *C-25* 6. GND (White-Black)
7. To Dash LED *C-17* A5 (Pink-White)
8. Green-Red to *C-30, C-63* & *C-82*
9. To Dash LED *C-17* A7 (Pink-Black)

5. LHS Speaker (Cabin)



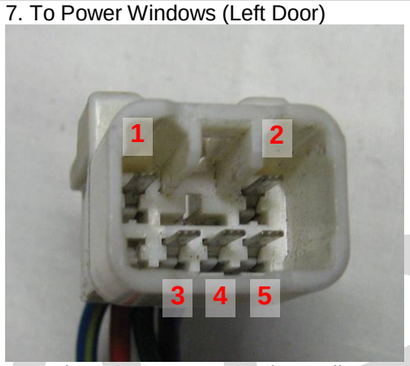
To *C-14*
Pink & Dark Pink

6. Interior Light Door Switch (Left Door)



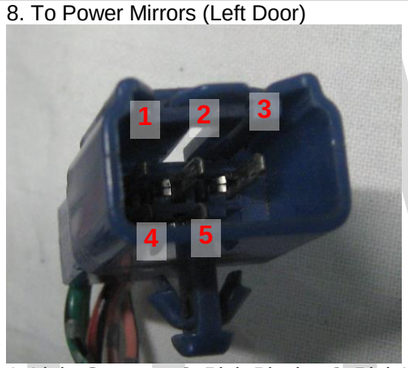
To Roof Mounted Light *C-9* (Red-White)

7. To Power Windows (Left Door)



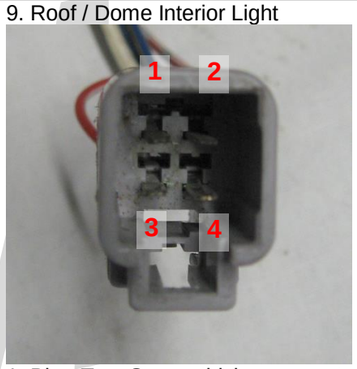
1. Red 2. Green 3. Blue-Yellow
4. Blue-Red 5. Blue-Green
1,2 to *C-22*
3,4,5 to *C-27*

8. To Power Mirrors (Left Door)



1. Light Green 2. Pink-Black 3. Pink-White
4. Green-Black 5. Red-Black
1,2,3 to *C-35*
4,5, to *C-31*

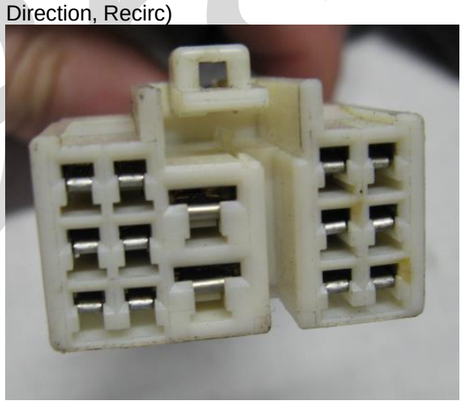
9. Roof / Dome Interior Light



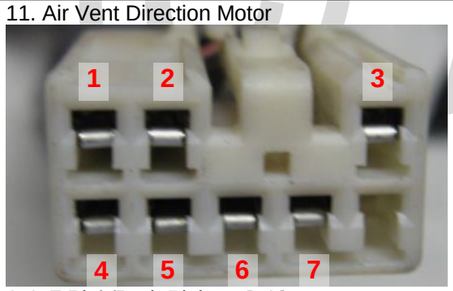
1. Blue-Two Green, thicker gauge to *C*-27
2. White-Black, thicker gauge GND
3. To *C-6* (Blue-Yellow)
4. From 'Dome' Fuse (Red-White)

1 & 2 not connected (for sunroof?)

10. Heater & A/C Dash Controls (Fan, Vent Direction, Recirc)

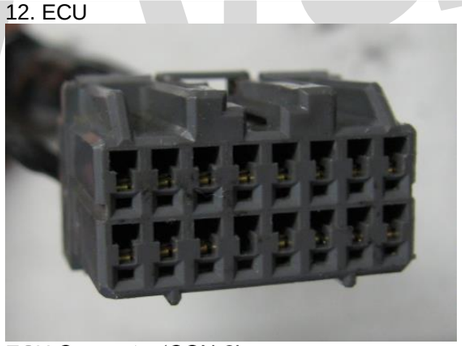


11. Air Vent Direction Motor



1-4, 7 Pink/Dark Pink to *C-10*
5. Red-Blue to 'Dash' Fuse
6 White-Black GND

12. ECU

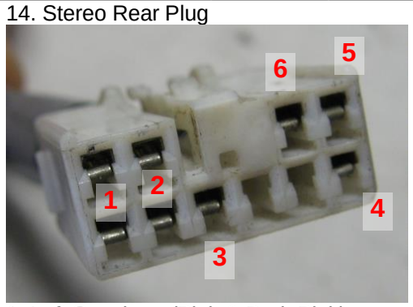


ECU Connector 'CON-2'

13. Cigarette Lighter Wires (no connectors shown)

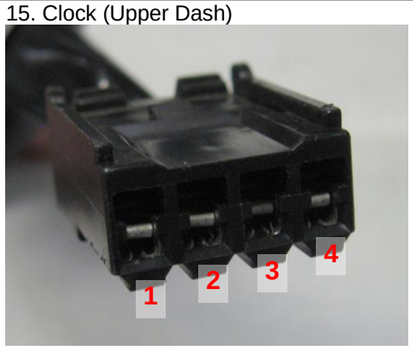
GND (White-Black)
12V Acc (Grey) From 'Radio' Fuse

14. Stereo Rear Plug



1. Left Speakers (Pink & Dark Pink)
2. Right Speakers (Light Green & Blue)
3. GND (Brown)
4. Park lights lighting signal (Green)
5. 12V Constant from 'Dome' Fuse (Blue-Yellow)
6. 12V Acc From 'Radio' Fuse (Grey)

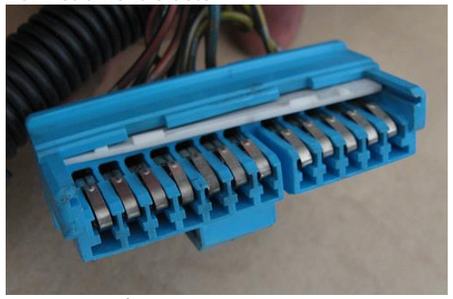
15. Clock (Upper Dash)



1. Park lights lighting signal (Green)
2. 12V Constant from 'Dome' Fuse (Blue-Yellow)
3. GND (White-Black)
4. 12V Acc (Grey) From 'Radio' Fuse

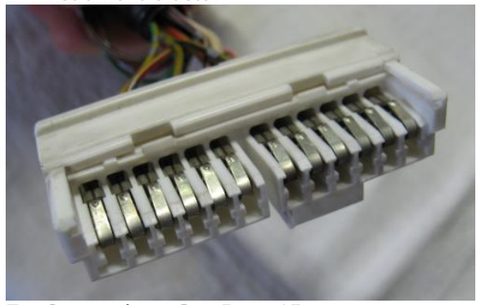
(c) Alan Vaughan

16. Instrument Cluster



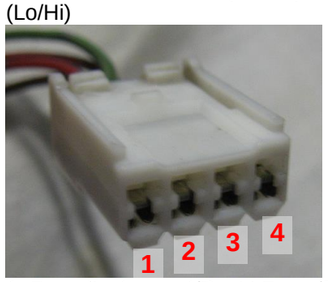
For Connections, See Page 15

17. Instrument Cluster



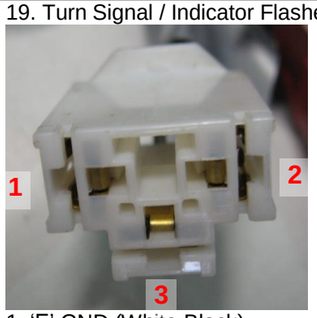
For Connections, See Page 15

18. Instrument Cluster (Boost pressure Gauge (Lo/Hi)



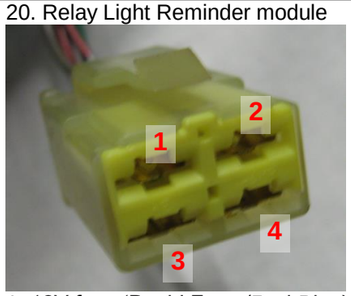
1. From 'Instrument/Dash' Fuse (Red-Blue)
2. Hi/Lo Boost Mode *C-33* (Light Green-Red)
3. Boost Press Signal *C-59* (Light Green)
4. GND (Brown)

19. Turn Signal / Indicator Flasher module



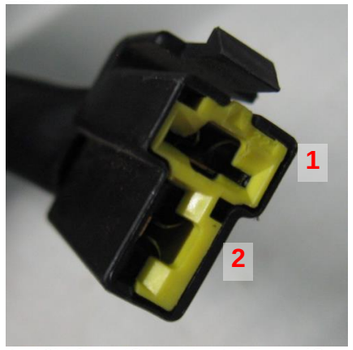
1. 'E' GND (White-Black)
2. 'L' from 'F' *C-32* (Light Green)
3. 'B' from 'TB' Turn Signals *C-32 & C-39* (Green-Blue)

20. Relay Light Reminder module



1. 12V from 'Dash' Fuse (Red-Blue)
2. 12V Acc from 'Radio' Fuse (Grey)
3. Park lights lighting signal (Green)
4. Door Switch *C-29* (Red-Green)

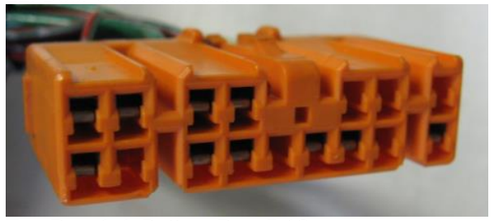
21. Brake Pedal



1. 12V from 'Stop Lights' Fuse (Green-Red)
2. To rear of car via *C-30* (Green-White)

Pedal Switch N/C (Normally Closed)

22. Door Control module

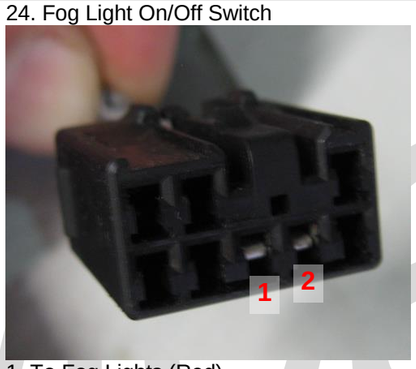


Top Row 1-4 Bottom Row 5-10
1. Red-Blue to 'Dash' Fuse 2. Red-Green
3. Red (Thicker gauge) 4. Green (Thick gauge)
5. From Power Window Fuse (Blue Thick gauge)
6. Green-Red 7. Red-White
8. Red-Green (From 'Dash' Fuse via Diode)
9. Green (Thick gauge)
10. GND (White-Black Thick gauge)
2 to Interior Light
3,4,9 to *C-7*
3,6,7,9 to *C-27*

23. RHS Speaker (Cabin)

To *C-14*
Light Green & Blue

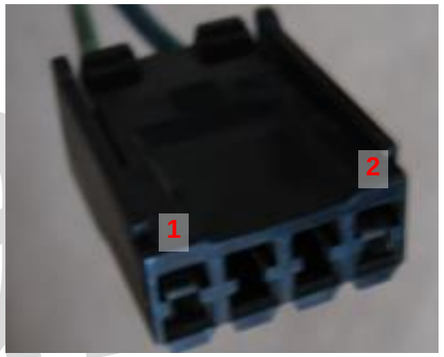
24. Fog Light On/Off Switch



1. To Fog Lights (Red)
2. 'T' *C-39* (Green)

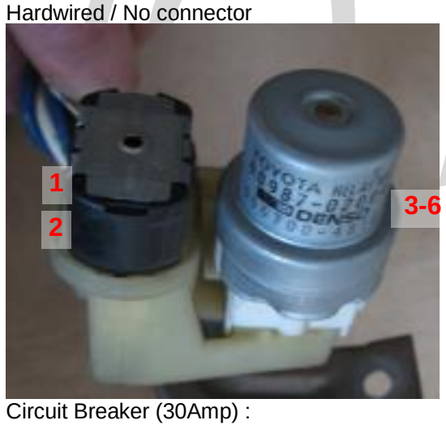
Switch N/O (Normally Open)

25. TEMS Switch



1. *C-4* Light Green
2. From 'Ignition' Fuse (Blue-Green)

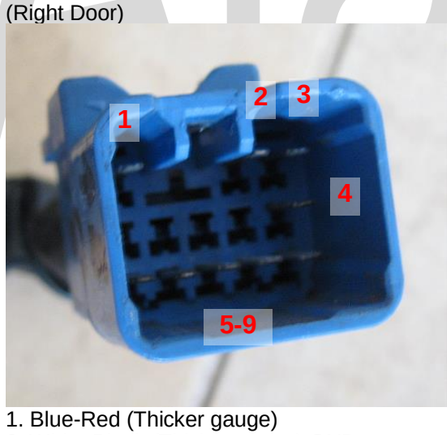
26. Circuit Breaker & Relay for power windows Hardwired / No connector



Circuit Breaker (30Amp) :
1. Ignition 'AM1' (White)
2. To *C-22* (Blue – Thicker gauge)

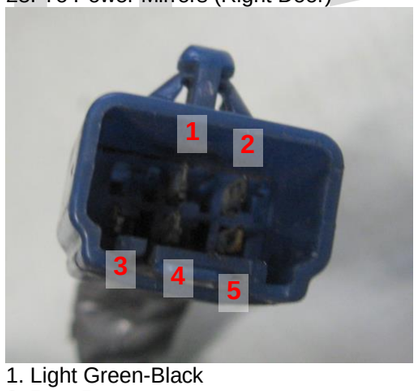
Relay :
3. From *C-22* (Blue-Green)
4. GND (White-Black)
Relay Contact Wires :
5. From *C-22* (Blue – Thick gauge)
6. From C-27 (Blue-Green – Thick gauge)

27. To Power Window / Sunroof switches (Right Door)



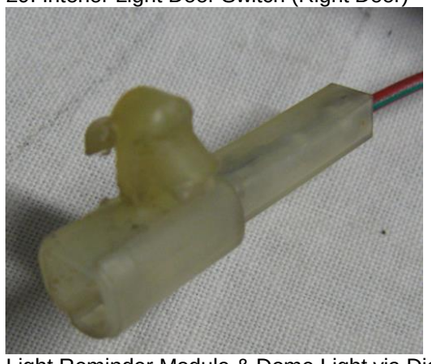
1. Blue-Red (Thicker gauge)
2. White-Black (Thicker gauge) GND
3. Green (Thicker gauge)
4. Blue-Yellow (Thicker gauge)
5. Green-Red
6. Red-White
7. Red (Thicker gauge)
8. Green (Thicker gauge)
9. Blue-Two Green (Thicker gauge) – Sunroof?
1,4,6,7,8 to *C-7*; 3,5,6,7,8 to *C-22*; 9 to *C-9*

28. To Power Mirrors (Right Door)



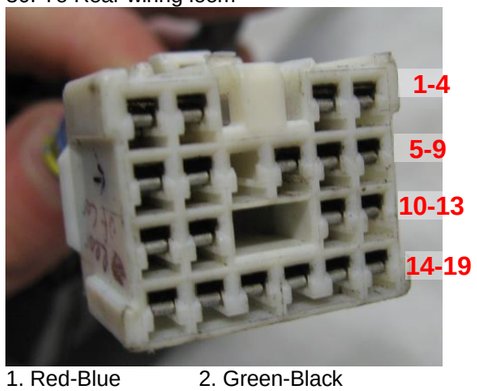
1. Light Green-Black
2. Blue-Black
3. Light Green-Red
4. Light Green-Black
5. Light Green
1,2 to *C-31*
3,4,5 to C-*35*

29. Interior Light Door Switch (Right Door)



Light Reminder Module & Dome Light via Diode
To *C-20*
(Red-Green)

30. To Rear wiring loom



1. Red-Blue 2. Green-Black
3. Green-Yellow 4. Green-Yellow
5. Red-Yellow 6. Yellow-Red
7. Black 8. Green
9. Green-Red 2 Dot 10. Blue-White
11. Green-White 12. Red-Black
13. Blue-Red 14. Yellow-Black
15. Yellow-Blue 16. Blue
17. Blue-Black 18. Blue-Yellow
19. Green-red 1 Dot

For Connections, See Page 11

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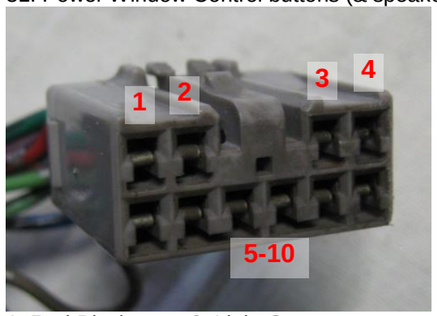
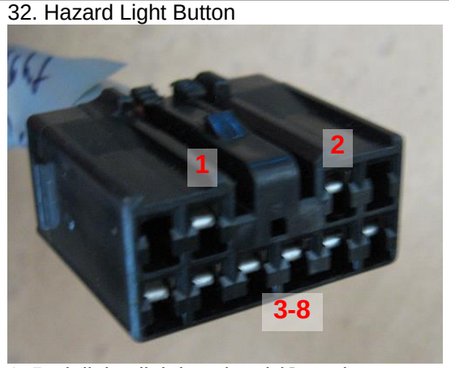
31. Power Window Control buttons (& speakers)



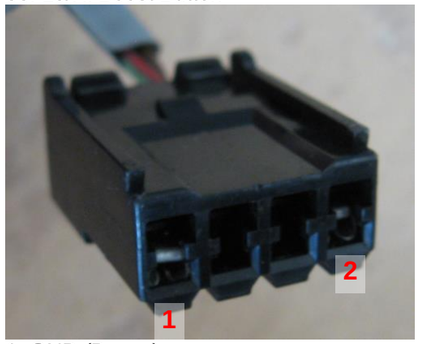
1. Red-Black 2. Light Green
3. Blue 4. Light Green-Black
5. Green-Black 6. Pink
7. Dark Pink 8. From 'Radio' Fuse (Grey)
9. Blue-Yellow 10. Blue-Black
2,3 to Left Door speaker
6,7 to Right Door Speaker
1,4,5,8,9,10 for Power Windows

32. Hazard Light Button



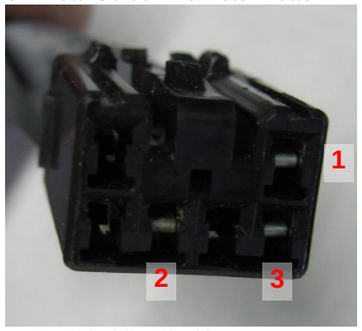
1. Park lights lighting signal (Green)
2. GND (White-Black)
3. 'TL' (Green-Black)
4. 'TR' (Green-Yellow)
5. 'F' (Light Green)
6. 'B2' From 'Haz/Horn' Fuse (Green-White)
7. 'TB' (Green-Blue)
8. 'B1' From 'Turn Signals' Fuse (Red-Blue)

33. Lo/Hi Boost Button



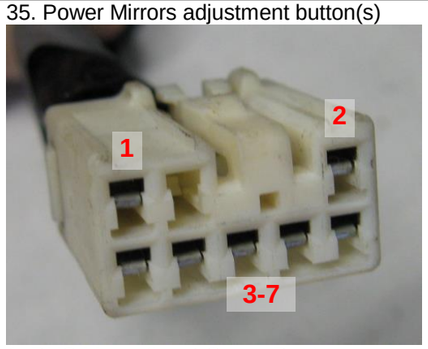
1. GND (Brown)
2. ECU 'BC' (Light Green-Red) & Dash boost indicator via *C-18*

34. Rear Screen Demister Button



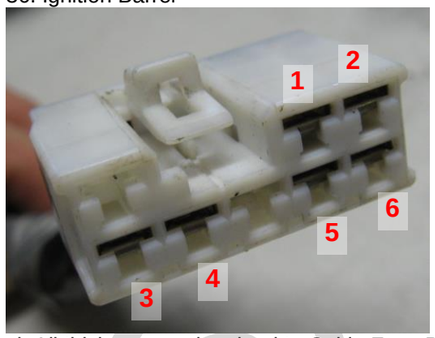
1. GND (White-Black)
2. From 'Rear Demister' Fuse (Light Green)
3. To Demister via *C-30* (Black)

35. Power Mirrors adjustment button(s)



1. RHS Mirror (Light Green-Red)
2. GND (White-Black)
3. From 'Radio' Fuse (Grey)
4. RHS Mirror (Light Green-Black)
5. LHS Mirror (Pink-Black)
6. LHS & RHS Mirror (Light Green)
7. LHS Mirror (Pink-White)

36. Ignition Barrel

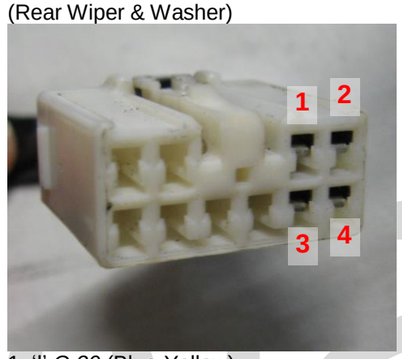


sigAll thick gauge wires lead to Cabin Fuse Box
1. 'AM2' (White-Red) 2. 'IG2' (Black-Red)
3. 'IG1' (Black-Yellow) 4. 'ACC' (Blue-Red)
5. 'AM1' (White)
6. 'ST1' ECU 'NSW' & ECU 'STA' (Black-White)

Missing pin (top left – Not Connected) is 'ST2'

See Page 11 for Conductions

37. Steering column combination switch (Rear Wiper & Washer)



1. 'I' *C-30* (Blue-Yellow)
2. 'E' GND (White-Black)
3. '+1' *C-30* (Blue-Black)
4. 'W' *C-95* (Blue-White)

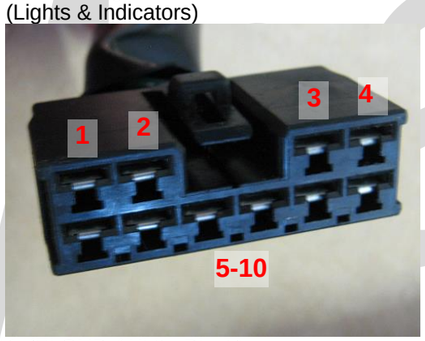
See Page 11 for Conductions

38. Steering column combination switch (Front Wipers & Washer)

1. '+1' *C-49* (Blue-Black)
2. '+2' *C-49* (Blue-Orange)
3. 'EW' GND (White-Black)
4. 'W' *C-95* Washer (White-Blue)
5. '+B' *C-49* (Blue) From 'Wipers' Fuse
6. '+S' *C-49* (Blue-Yellow)

See Page 11 for Conductions

39. Steering column combination switch (Lights & Indicators)

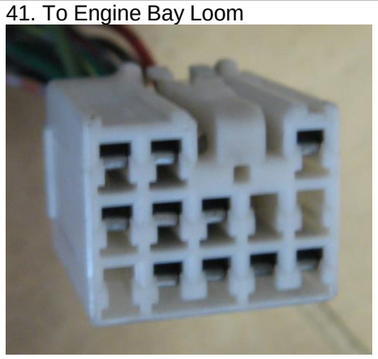


1. 'E' GND (White-Black)
2. 'TL' Turn Signals Left (Green-Black)
3. 'TB' Turn Signals 12V (Green-Blue)
4. 'TR' Turn Signals Right Green-Yellow (1 Dot)
5. 'HU' Headlights High Beam (Red-Yellow)
6. 'RF' ECU 'ELS' (Red-Green)
7. 'HL' Headlights Low Beam (Red-White)
8. 'TI' To Park Lights (Green)
9. 'BO' From 'Tail Lights' Fuse (Green-White)
10. 'HO' Horn (Green-Yellow 2 Dot) – Horn button shorts this pin to GND
See Page 11 for Conductions

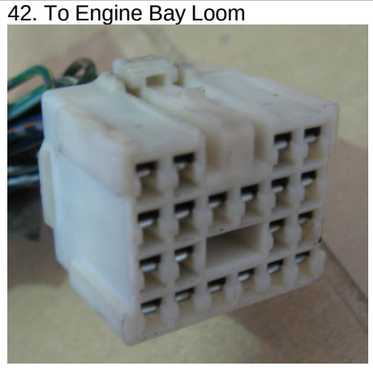
40. Foot/Lower Air Direction Motor

1. Dark Pink-Green to *C-10*
2. Dark Pink-Red to *C-10*
3. Red-Blue to 'Dash' Fuse

41. To Engine Bay Loom

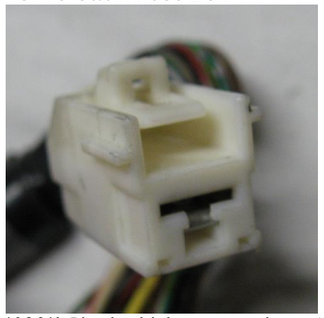


42. To Engine Bay Loom



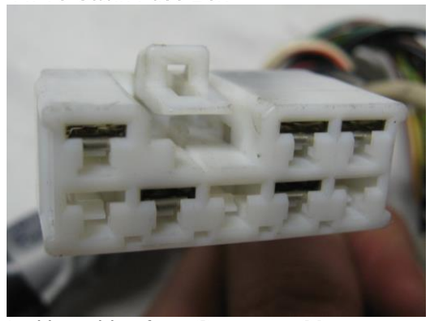
© Rave

43. To Cabin Fuse Box



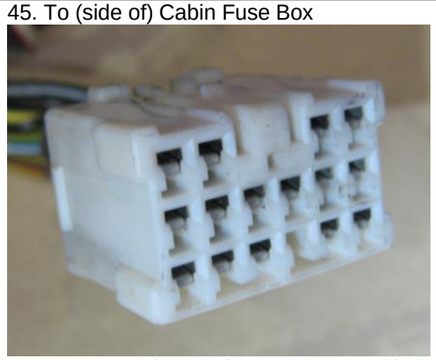
'AM1' Single thick gauge wire – 12V (White)

44. To Cabin Fuse Box



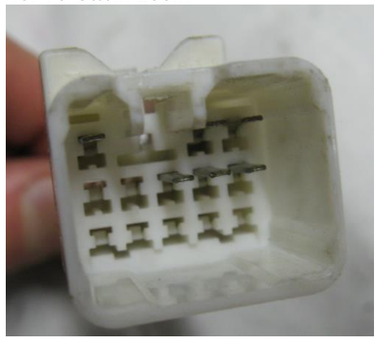
Ignition wiring from Connector 36

45. To (side of) Cabin Fuse Box

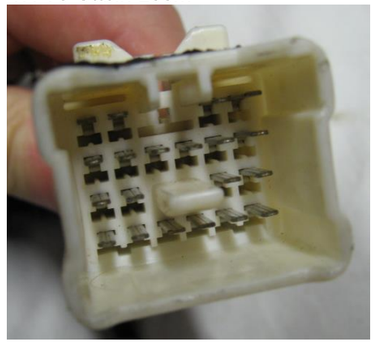


Bypasses fuse box and connects to Cabin loom

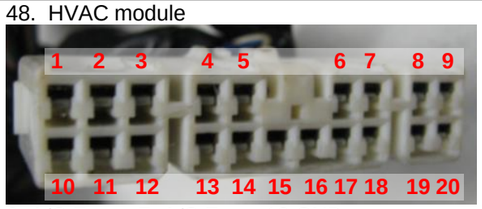
46. To Cabin Loom



47. To Cabin Loom



48. HVAC module

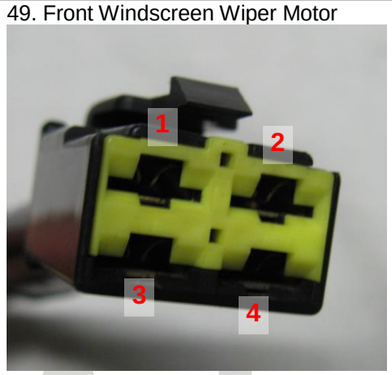


6. A/C Fuse 7. A/C Fuse 12.GND
13. ECU 'AC2' 14. ECU 'AC1' 15. ECU 'ACT'

Remaining wires connect to
C-51 *C-52* *C-68*
C-85 *C-90* *C-92*

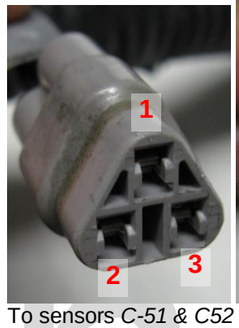
See above connector description for wire colours

49. Front Windscreen Wiper Motor

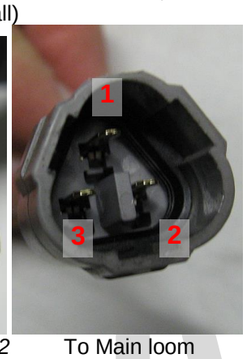


1. Blue-Orange (Fast wiper speed)
2. Blue-Black (Slow wiper speed)
3. Blue-Yellow (Intermittent/Park)
4. Blue (12V Constant - From 'Wipers' Fuse)
All wires to *C-38*. GND on wiper motor case.

50. Sub-harness connector to sensors (mounted above Ignitor on firewall)



To sensors *C-51 & C52*



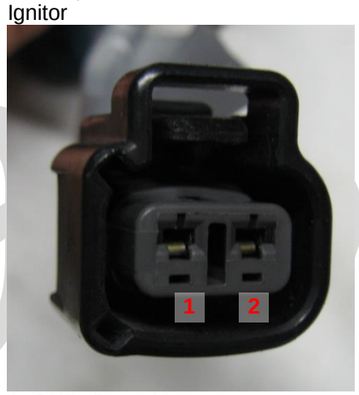
To Main loom

C48 Pin16 Green --- 1 --- Green-Blue *C-51*
C48 Pin17 Black ------ 2 --- Black-Yellow *C-51,C52*
C48 Pin18 Grey ------ 3 ----- Green-Red *C-52*

51. Vacuum Switch Valve
Used by HVAC module - RHS Sensor above Ignitor

1. GND (Black-Yellow)
2. To HVAC Module *C-48* (Green-Blue)

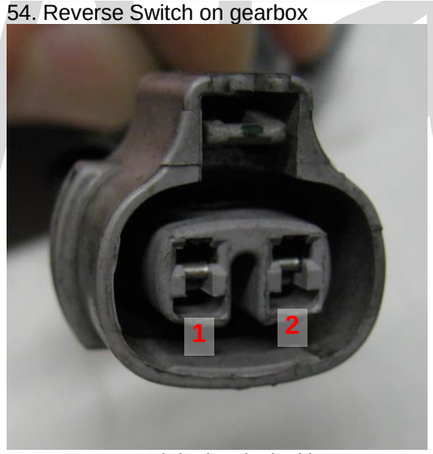
52. Idle Up Valve
Used by HVAC module - LHS Sensor above Ignitor



1. GND (Black-Yellow)
2. To HVAC Module *C-48* (Green-Red)

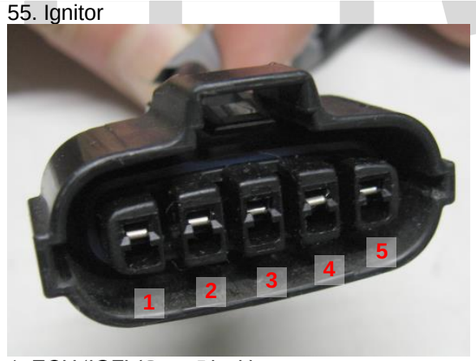
53. Diagnostics Port (hardwired – no connector)
For Connections, See Page 11

54. Reverse Switch on gearbox



1. To Reverse Light (Red-Black)
2. 12V from 'Instrument/Dash Cluster' fuse (Red-Blue)

55. Ignitor



1. ECU 'IGF' (Grey-Black)
2. ECU 'IGT' (Light Green-White)
3. 'IG2' Ignition (Black-Red)
4. Diagnostics Socket 'IG-' (Black)
5. Pin 2 on Coil *C-56* (Black-White)

56. Coil (Under Ignitor)

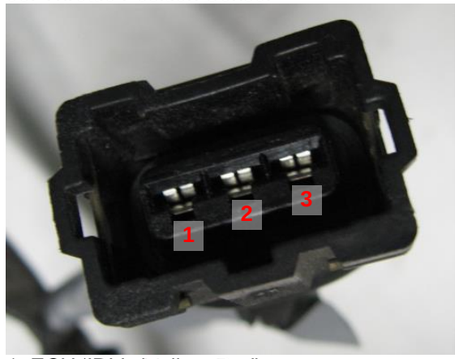
1. 'IG2' Ignition (Black-Red)
2. Pin 5 on Ignitor*C-55* (Black-White)

57. Unknown

1. 'IG2' Ignition (Black-Red)
2. GND (White-Black)

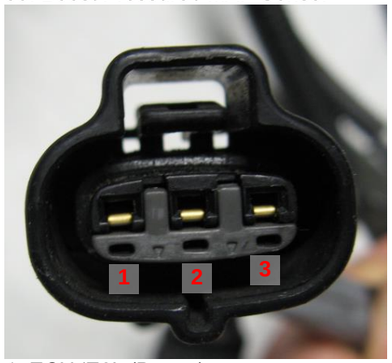
© Ian Wa

58. Throttle Position Sensor



1. ECU 'IDL' (Yellow-Red)
2. ECU 'E2" (GND - Brown)
3. ECU 'PSW' (Yellow-Green)

59. Boost Pressure / MAP Sensor

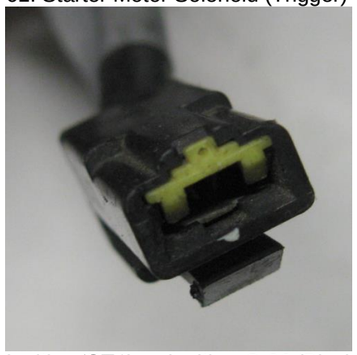


1. ECU 'E2' (Brown)
2. ECU 'PIM' (Green)
3. ECU 'VCC' (Green-Red)

60. Starter Motor (Main Voltage)

Single black wire (heavy gauge) – connected directly to Battery

61. Starter Motor Solenoid (Trigger)

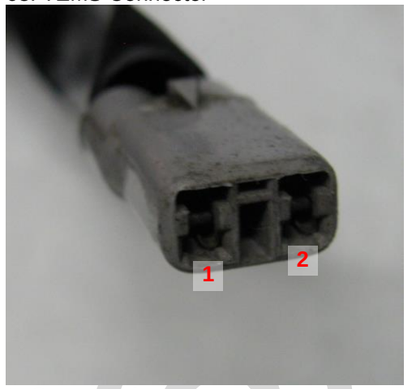


Ignition 'ST1' on Ignition *C-36* (Black-White)

62. Alternator

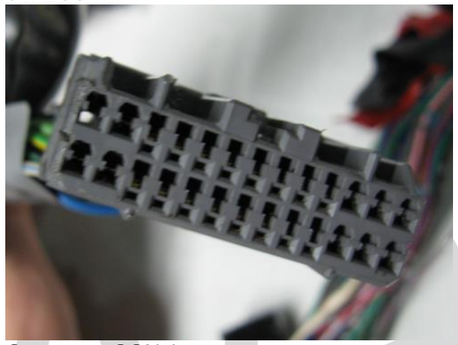
1. From 'Instrument/Dash Cluster' Fuse (Red-Black)
2. Connection to Battery via AM1 fuse (White)
3. To *C-17* A4 Battery Light in instrument cluster (Yellow-Black)
4. Connection to Battery via AM1 fuse (White)

63. TEMS Connector



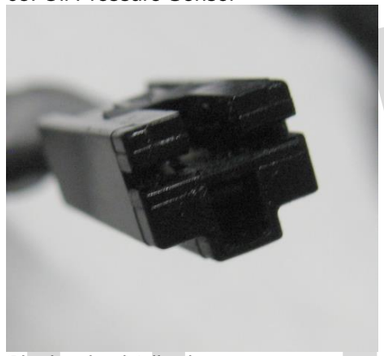
1. Green-Red
2. Green-Yellow
(To Rear of Car *C-30*)

64. ECU



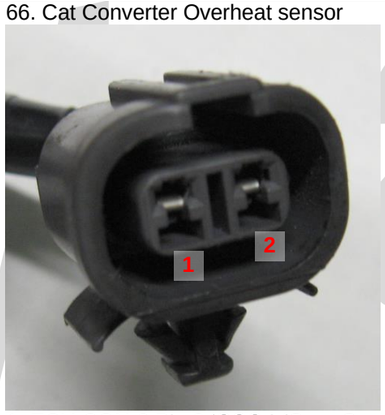
Connector CON-1

65. Oil Pressure Sensor



Single wire (Yellow)
Connects to Instrument cluster *C-17* A8

66. Cat Converter Overheat sensor

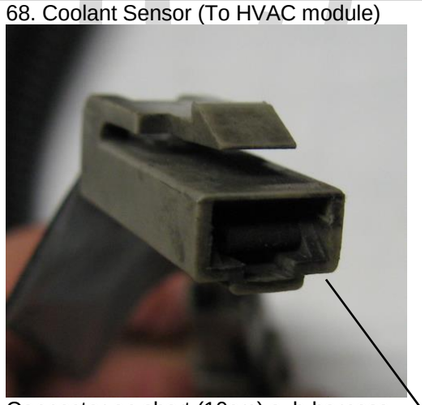


1. ECU CON-2 Pin 3 'CCO ' (Grey-Green)
2. Ground (Brown)

67. Oxygen Sensor

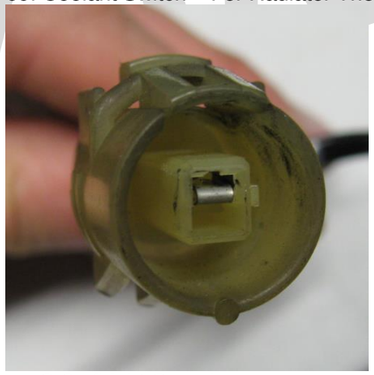
ECU 'OX' (White)

68. Coolant Sensor (To HVAC module)



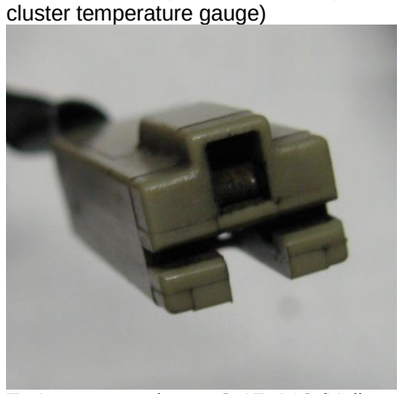
Connector on short (10cm) sub-harness
Wire in White in sub-harness;
Blue-Yellow in main loom
To HVAC module *C-48 Pin 11*

69. Coolant Switch – For Radiator Thermofan



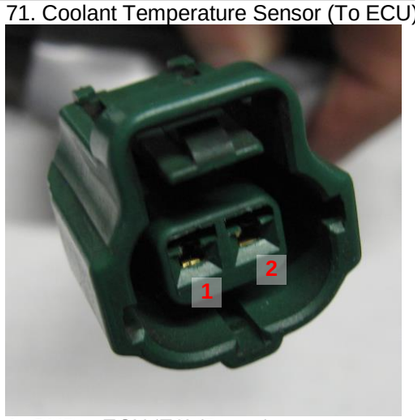
To *C-85* (Light Green)
Temperature Switch Normally Open (N/O)

70. Coolant Temperature Sensor (To Instrument cluster temperature gauge)



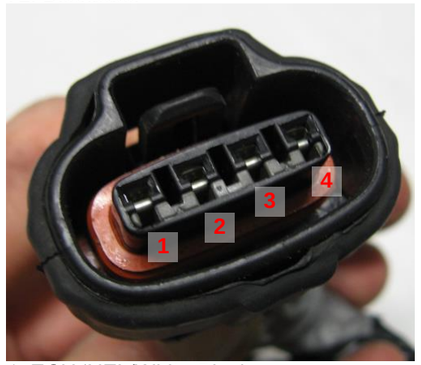
To Instrument cluster *C-17* A12 (Yellow-Green)

71. Coolant Temperature Sensor (To ECU)



1. GND – ECU 'E2' (Brown)
2. ECU 'THW' (Green-Black)

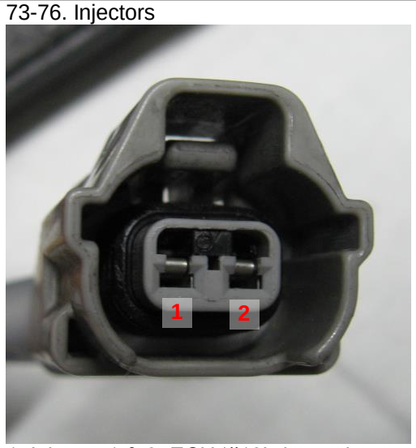
72. Distributor



1. ECU 'NE' (White-Blue)
2. ECU 'G1' (Orange-Blue)
3. ECU 'G2' (Blue-Green)
4. ECU 'G-' (Blue-Yellow)

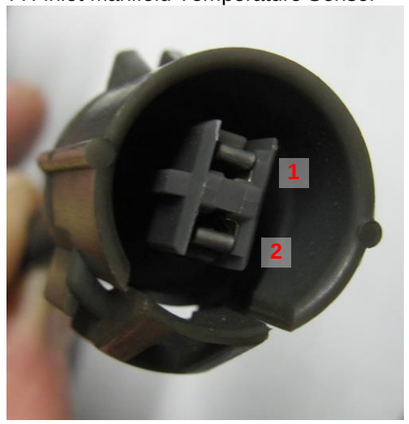
Draft Version

73-76. Injectors



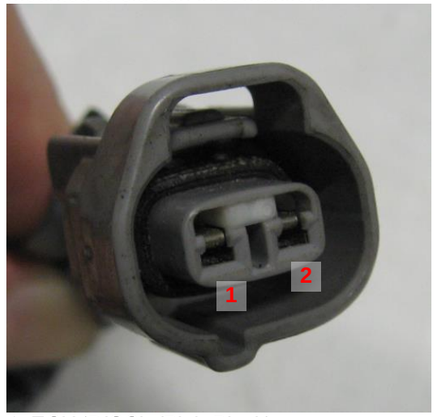
1. Injector 1 & 3 ECU '#10' (Green)
   Injector 2 & 4 ECU '#20' (Yellow)
2. Ignition IG2 (Black-Red)

77. Inlet Manifold Temperature Sensor



1. Sensor GND ECU 'E2' (Brown)
2. ECU 'THA' (Light Green-Black)

78. Idle Up Valve
Mounted to side on inlet manifold



1. ECU 'DISC' (Pink-Black)
2. 12V via EFI Relay (Black-Red)

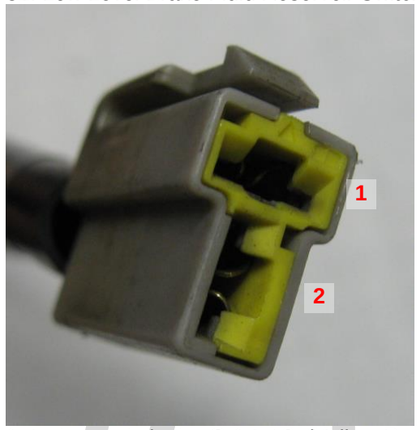
79. Vacuum Solenoid Valve
Controls pressure to wastegate actuator

1. 12V via EFI Relay (Black-Red)
2. ECU 'VSV1' (Pink-White)

80. Knock Sensor
Located at rear of block

ECU 'KNK' (Black)

81. Low Level Brake Fluid Reservoir Switch

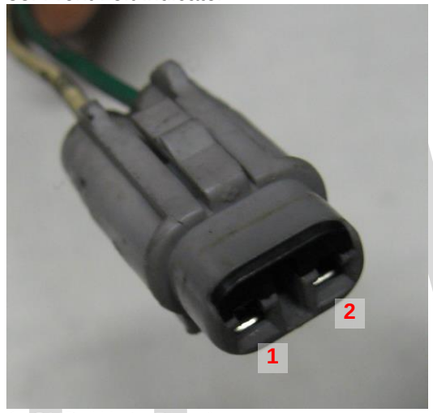


1. Instrument cluster *C-17* A3 (Yellow-Red)
2. GND (White-Black)
N/O circuit. At low fluid level, circuit is shorted.

82. TEMS Connector

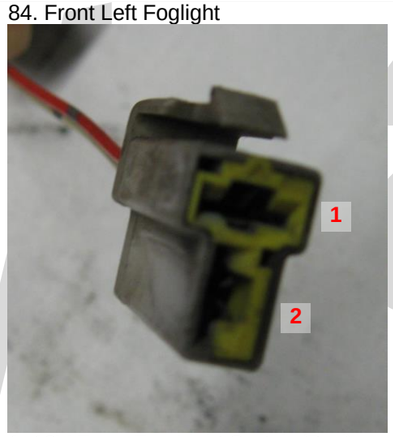
1. Green-Yellow
2. Green-Red

83. Front Left Indicator



1. GND (White-Black)
2. 'TL' (Green-Black)

84. Front Left Foglight



1. 12V From Fog Light Switch (Red)
2. GND (White-Black)

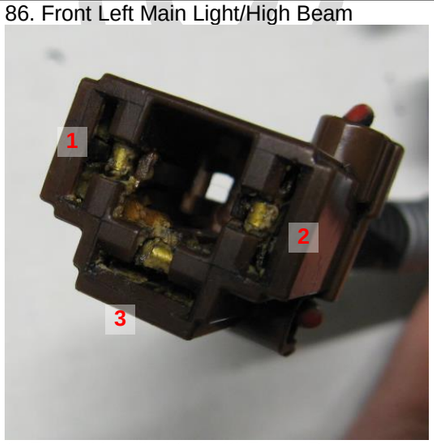
85. A/C Pressure Switch

Mounted in A/C Piping (no connectors shown).

Switch Normally Closed (N/C)

Light Green – *C-69* (Coolant Sensor)
Light Green-Red – to 'Cooling Fan' relay, 'A/C Fan No. 2" relay & HVAC Module *C-48*

86. Front Left Main Light/High Beam



1. From 'Headlights Left' Fuse (Red)
2. 'HU' High Beam (Red-Yellow)
3. 'HL' Low Beam (Red-White)

87. Front Left Park Light

1. GND (White-Black)
2. Lights Signal (Green)

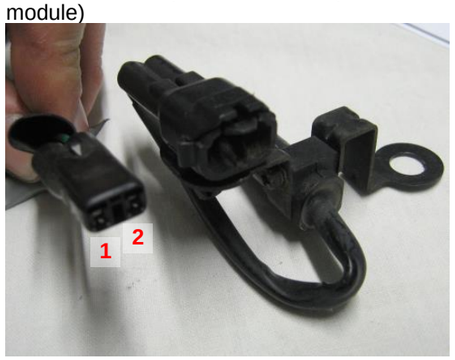
88. Radiator Fan

1. GND (White-Black)
2. 12V from 'Cooling Fan' or 'A/C Fan No.2" Relay (Blue-Black)

89. A/C Condenser Fan

1. To 'A/C Fan No. 2' relay (Blue-White)
2. To 'Cooler Fan' relay (Blue)

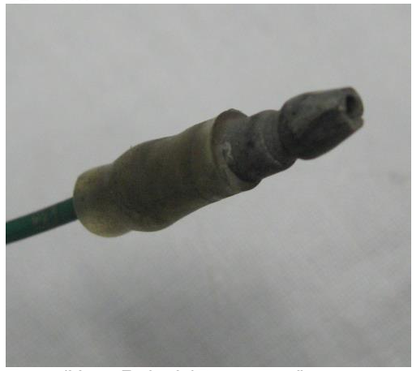
90. Ambient Temperature Sensor (To HVAC module)



1. White-Green to *C-48 Pin 20*
2. Green-Yellow to *C-48 Pin 19*

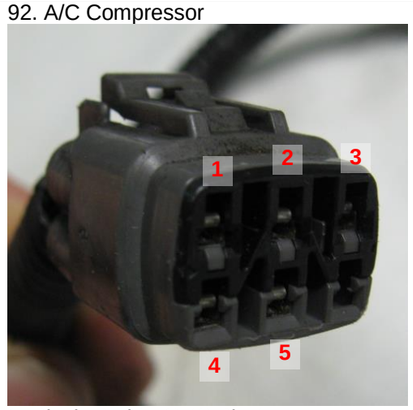
Alfa Romeo Cars

91. Horn



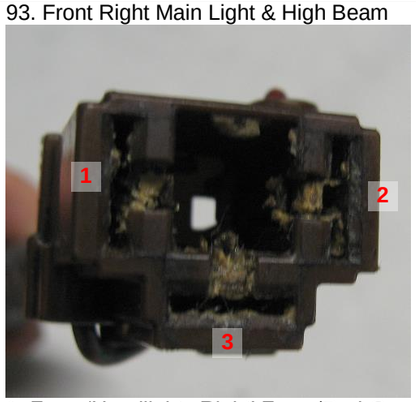
From ‘Horn Relay’ (Green-Red)

92. A/C Compressor



1. Black-Red to *C-48 Pin 4*
2. Blue-Black to *C-48 Pin 5 & A/C Relay #2 Pin 2*
3. Green-Red to *C-48 Pin 9*
4. Yellow-Red to *C-48 Pin 2*
5. White-Yellow to *C-48 Pin 3*

93. Front Right Main Light & High Beam

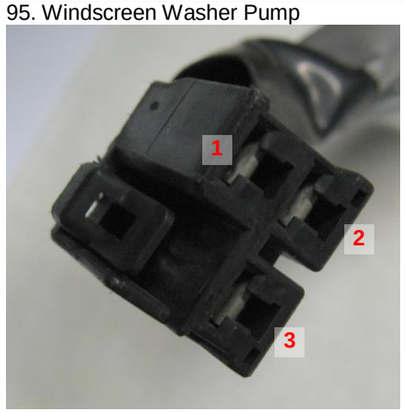


1. From ‘Headlights Right’ Fuse (Red-Green)
2. ‘HU’ High Beam (Red-Yellow)
3. ’ HL’ Low Beam (Red-White)

94. Front Right Park Light

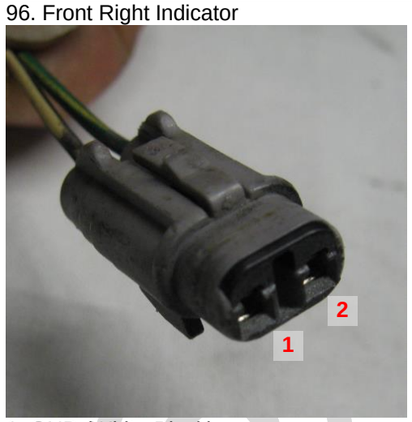
1. GND (White-Black)
2. Lights Signal (Green)

95. Windscreen Washer Pump



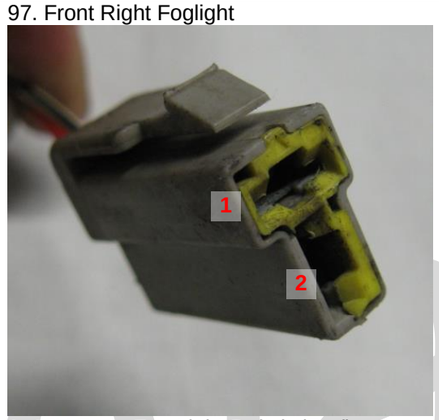
1. Rear Washer To *C-37* (Blue-White)
2. Front Washer To *C-38* (White-Blue)
3. 12V Constant - From ‘Wipers’ Fuse (Blue)

96. Front Right Indicator



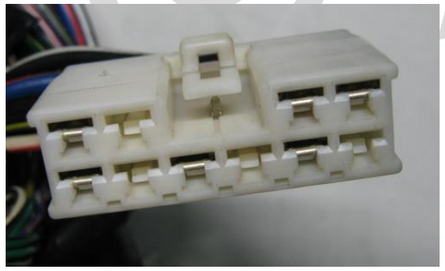
1. GND (White-Black)
2. ‘TR’ (Green-Yellow)

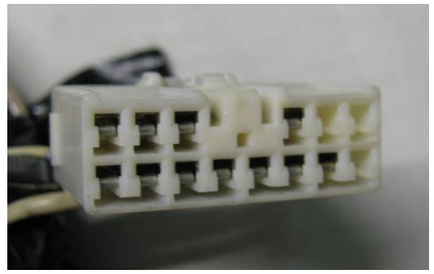
97. Front Right Foglight

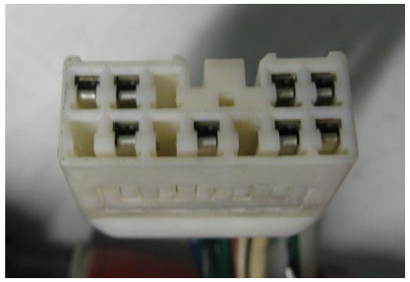


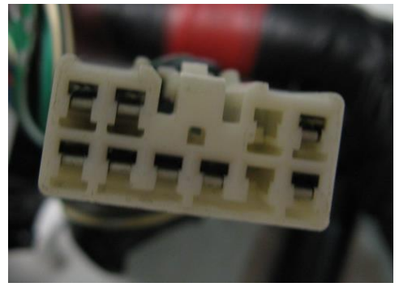
1. 12V From Fog Light Switch (Red)
2. GND (White-Black)

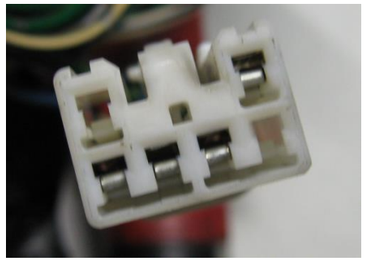
Connectors plugged into the base of Cream-coloured Cabin Fuse Box











(c) Alan Mears

Conduction Code Legend :

B1	Battery Source 1
B2	Battery Source 2
F	Flasher
TB	Turn Signal Battery / Power
TL	Turn Left Signals
TR	Turn Right Signals
B / BI / +B	Battery (12V)
E / EW	Earth/Ground
HL	Headlight – Low Beam
HU	Headlight – Upper/High Beam
RF	Signal to ECU ('ELS')
TI	Taillights (and front Park lights)
+S	Wiper Park/Intermittent
+1	Slow/Low Wiper Speed
+2	Fast/High Wiper Speed
W	Washer Pump

Diagnostics Connector 53 :

Pin	Wire Colour	Connection / Function
F_P	Blue-White	12V to Fuel Pump
CC_O	Grey-Green	ECU 'CCO'
+B	Black-Red	12V (via EFI relay)
W	N/A	Not Connected
T_{E1}	Green-Blue	ECU 'T"
T_{E2}	N/A	Not Connected
V_{F1}	Green-Yellow	ECU 'VF'
V_{F2}	N/A	Not Connected
E_1	Brown	ECU 'E1' (Ground)
Ox_1	White	ECU 'OX' (Oxygen Sensor)
Ox_2	N/A	Not Connected
IG-	Black	Ignitor Connector 55, Pin 4
AB	N/A	Not Connected
CC_2	N/A	Not Connected
T_S	Red-Blue	Not Connected
O_{PT}	Red-Yellow	To HVAC module, C-48 Pin 1
T_C	Red-Black	Not Connected
T_T	N/A	Not Connected

Connector 30 :

Pin	Wire Colour	Function
1	Red-Blue	From 'Instrument/Dash' Fuse
2	Green-Black	Left Indicator 'TL'
3	Green-Yellow	Right Indicator 'TR'
4	Green-Yellow	TEMS
5	Red-Yellow	Overdrive signal (AT Transmission)
6	Yellow-Red	Handbrake Ground
7	Black	Demister Power
8	Green	Rear Park Lights 'T'
9	Green-Red (2Dot)	TEMS
10	Blue-White	Fuel Pump
11	Green-White	Rear Brake Lights
12	Red-Black	Reverse Light
13	Blue-Red	Unknown
14	Yellow-Black	Low Fuel Warning Signal
15	Yellow-Blue	Fuel Level Signal
16	Blue	Rear wiper power
17	Blue-Black	Rear Wiper – Slow speed
18	Blue-Yellow	Rear Wiper – Intermittent
19	Green-Red (1Dot)	Unknown

O—O : Conduction between pins

Connector 32 : Hazard Switch

Pin ➔	8	6	5	7	3	4
Switch Position ↓	**B1**	**B2**	**F**	**TB**	**TL**	**TR**
OFF	O		O			
ON		O	O	O	O	O

Connector 36 : Ignition

Pin ➔	5	4	3	6	1	2
Key Position ↓	**AM1**	**ACC**	**IG1**	**ST1**	**AM2**	**IG2**
LOCK						
ACC	O	O				
ON	O	O	O		O	O
START	O		O	O	O	O

Connector 37 : Rear Wiper

Pin ➔	2	1	3	4
Switch Position ↓	**E**	**I**	**+1**	**W**
OFF				
INT	O	O		
ON	O		O	
WASH	O			O

Connector 38 : Front Wipers

Pin ➔	5	6	1	2	3	4
Switch Position ↓	**+B**	**+S**	**+1**	**+2**	**EW**	**W**
INT		O	O			
LOW	O		O			
HIGH	O			O		
WASH					O	O

Connector 39 : Lights & Indicators

Pin ➔		9	8	6	1	5	7
Switch Position ↓		**BI**	**TI**	**RF**	**E**	**HU**	**HL**
OFF							
	High Beam						
	Flash/Pass				O	O	
Park Lights		O	O				
	High Beam	O	O				
	Flash/Pass	O	O		O	O	
Main Lights		O	O	O	O		O
	High Beam	O	O	O	O	O	
	Flash/Pass	O	O	O	O	O	

Connector 39 : Turn Signals

Pin ➔	2	3	4
Switch Position ↓	**TL**	**TB**	**TR**
LEFT	O	O	
OFF			
RIGHT		O	O

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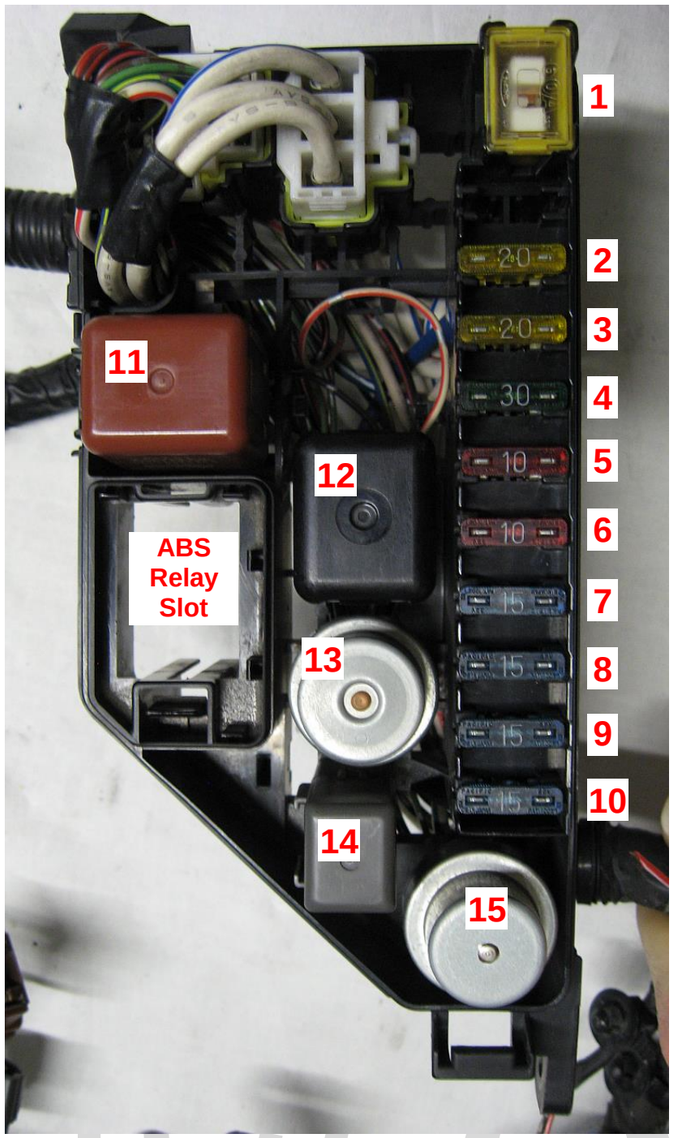
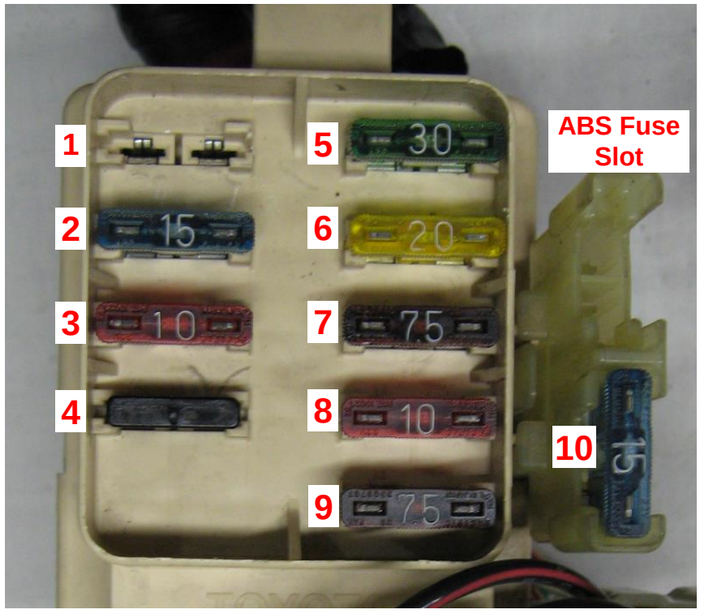


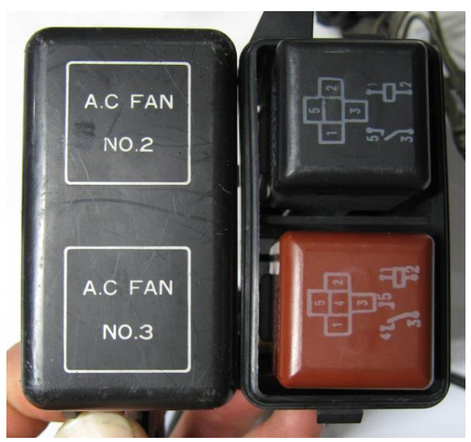
Engine Bay Fuse Box

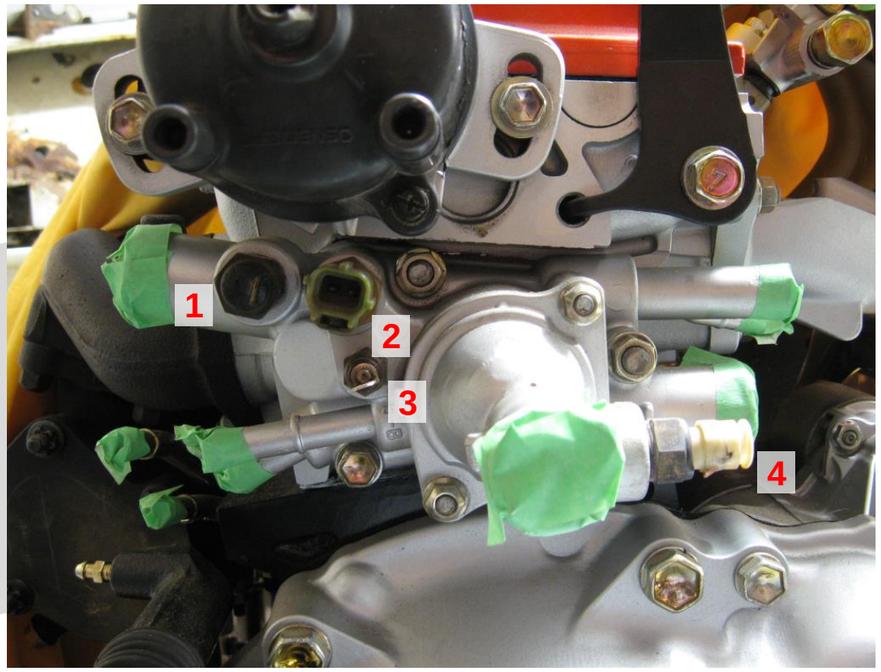
1	AM1	60A
2	A/C Condenser Fan (CDS)	20A
3	Radiator Fan	20A
4	Heater	30A
5	Headlights (Left)	10A
6	Headlights (Right)	10A
7	Interior Dome Light	20A
8	AM2	15A
9	Hazard/Horn	15A
10	EFI	15A
11	Main Relay	-
12	'Cooler' Relay	-
13	EFI Relay	-
14	Horn Relay	-
15	'Cooling Fan' Relay	-



Cabin Fuse Box

1	Radio	10A
2	Tail Lights	15A
3	Stop Lights	10A
4	Not Connected	-
5	Rear Demister	30A
6	Wipers	20A
7	Turn Signals	7.5A
8	A/C	10A
9	Instrument/Dash Cluster	7.5A
10	TEMS	15A





Sensors/Switches relating to coolant around thermostat housing :
1 – Connector 68. To HVAC Module
2 – Connector 71. To ECU
3 – Connector 70. To Dash Gauge
4 – Connector 69. Temperature Switch for Radiator Fan

Radiator Fan Switch (C-69) has internal switch :
Conduction when Coolant < 93°C
No Conduction when Coolant > 93°C

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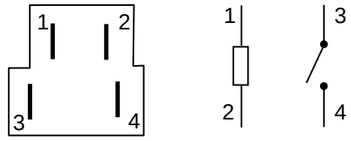
Relay Connection & Pin Details

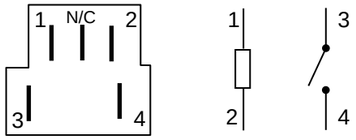
#11
Main Relay
(N/O)



1. Ignition ‘IG2’ (Black-Red)
2. GND (White-Black)
3. ‘AM1’ Fuse (White)
4. Radiator Fan Fuse (White-Red)

Part # 90987-02006

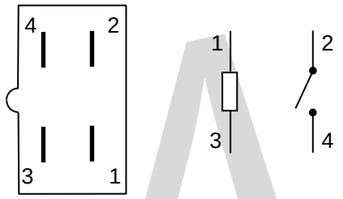
#12
‘Cooler’ Relay
(N/O)



1. Ignition ‘IG2’ (Black-Red)
2. GND (White-White)
3. *C-89* - A/C Fan (Blue)
4. CDS Fan Fuse (Blue-Black)

Part # 90987-04002

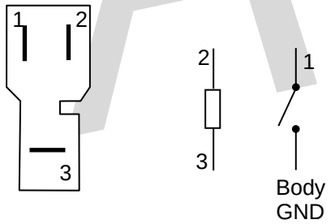
#13
EFI Relay
(N/O)
22 Amp rating



1. EFI Fuse (White-Red)
2. Ignition ‘IG2’ (Black-Red)
3. *C-1* – Circuit Opening Relay Pin 2 (Black-Red)
4. GND (White-Black)

Part # 90987-02004

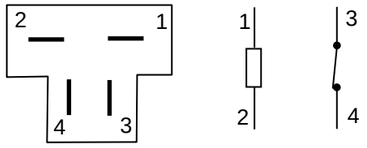
#14
Horn Relay
(N/O)



1. *C-91* – Horn (Green-Red)
2. Horn Fuse (Green-White)
3. *C-39* - Horn button (Green-Yellow)

Part # 86530-20070

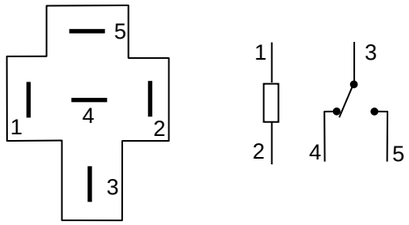
#15
‘Cooling Fan’ Relay
(N/C)
15 Amp rating



1. A/C Fan No. 3 Relay Pin 1 & *C-85* A/C Pressure Switch (Light Green-Red)
2. Ignition ‘IG2’ (Black-Red)
3. Radiator Fan Fuse (White-Green)
4. A/C Fan No. 3 Relay Pin 5 & Radiator Fan (Blue-Black)

N/C Relay - *Pins 3 & 4 are shorted when coil is not energized*

A/C Fan
No.3 Relay

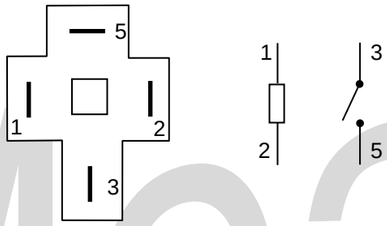


1. ‘Cooling Fan’ Relay Pin 1 & *C-85* A/C Pressure Switch (Light Green-Red)
2. Ignition ‘IG2’ (Black-Red)
3. A/C Fan No. 2 Relay Pin 3 (Blue-Red)
4. GND (White-Black)
5. Radiator Fan (Blue-Black)

Pins 3 & 4 are connected when coil is not energized

Part # 90987-02010

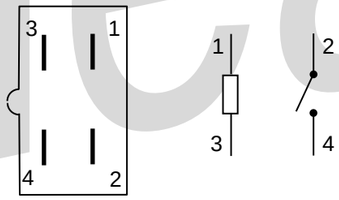
A/C Fan
No.2 Relay
(N/O)



1. GND (White-Black)
2. *C-48* – HVAC & *C-92* (Black-Blue)
3. A/C Fan No. 3 Relay Pin 3 (Blue-Red)
5. A/C Condenser Fan (Blue-White)

Part # 90987-04003

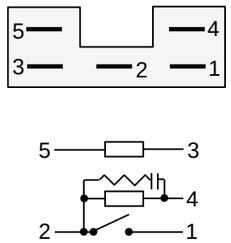
Power Windows
Relay
(N/O)
22 Amp rating
(Connector 26)



1. *C-22* Door Control Module (Red-Green)
2. From Power Windows Fuse (Blue)
3. GND (White-Black)
4. To Power Window (Blue-Green)

Part # 90987-02004

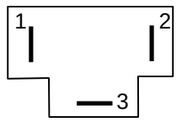
Circuit Opening
Relay
(N/O)
(Connector 1)



1. To Fuel Pump (Blue-White)
2. 12V from ‘EFI’ Relay & ECU ‘BATT’ (Black-Red)
3. From Ignition ‘ST1’ & To Starter Motor solenoid & ECU ‘STA’ (Black)
4. To ECU ‘FC’ Fuel Cut (Green)
5. GND (White-Black)

Part # 85910-10010

Flasher Relay
(Connector 19)



1. ‘F’ Flasher from Hazard Switch (Light Green)
2. GND (White-Black)
3. ‘TB’ Turn Signals (Green-Blue)

Part # 81980-12070

N.B. Diagrams are of relay plugs as viewed from above fuse box or at connector, not underside of actual relays

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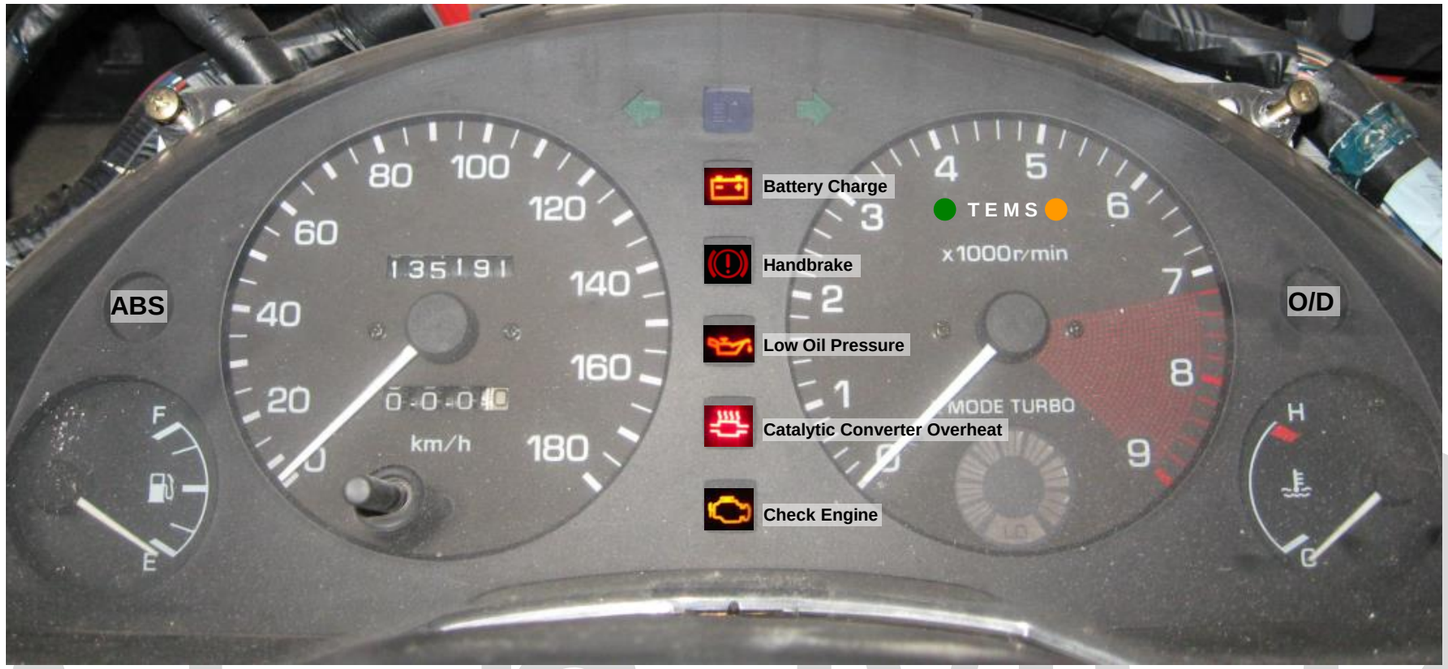
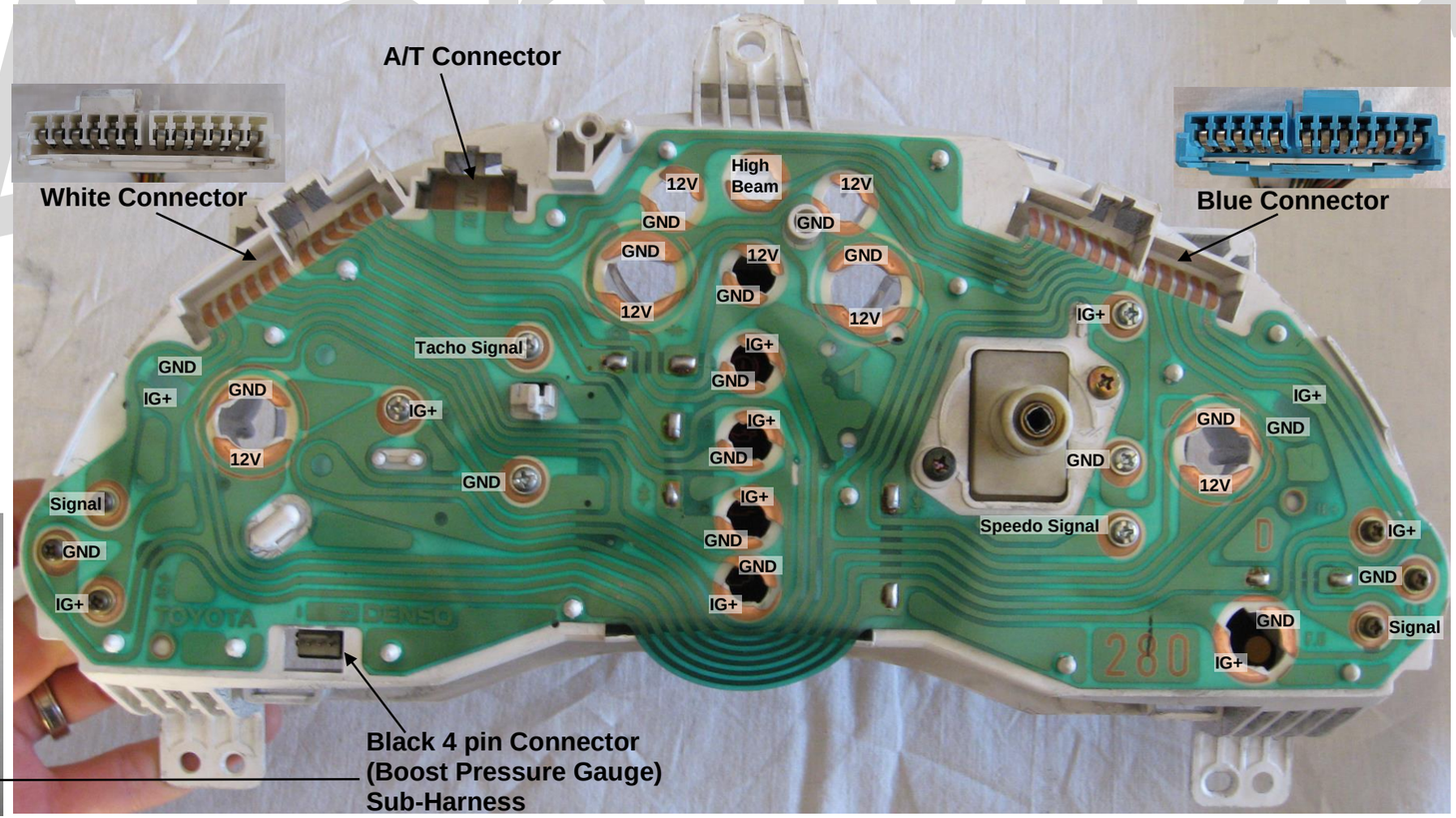
Instrument/Gauge cluster layout & PCB

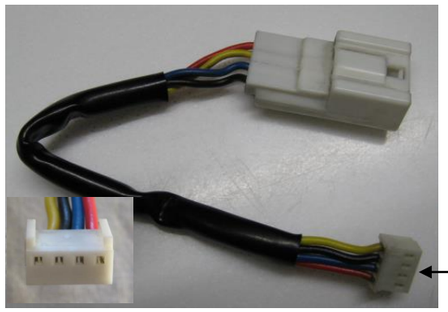


Note : photos are from 1991 (Gen 1) M/T 4E-FTE with no ABS or TEMS

ABS, TEMS & A/T Overdrive LED/light locations have been indicated



Boost Pressure Gauge Sub-Harness (connects between instrument cluster & cabin wiring loom C-18)



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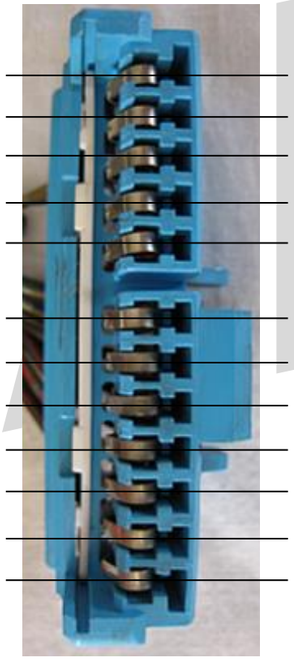
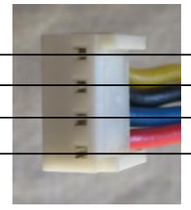
Instrument/Gauge cluster wiring

Lights (Positive)	A1	Green 'T'
Battery Charge (Positive)	A2	Black-Red 'IG2'
Handbrake Light (Ground)	A3	Yellow-Red
Battery Light (Ground) from Alternator	A4	Mustard-Black (1 Silver dot on wire)
TEMS 'Normal' (Power)	A5	Pink-White
Tacho Signal (IG-) (Ground)	A6	White (Ignitor Pin 4)
TEMS 'Sport' (Power)	A7	Pink-Black
Oil Pressure Light (Ground)	A8	Mustard
Cat Converter Overheat (Ground)	A9	Grey-Black (ECU Pin 'EGW')
Engine Check Light (Ground)	A10	Green-White (ECU Pin 'W')
O/D OFF (Ground)	A11	Red-Yellow
Temperature Gauge Signal	A12	Yellow-Green

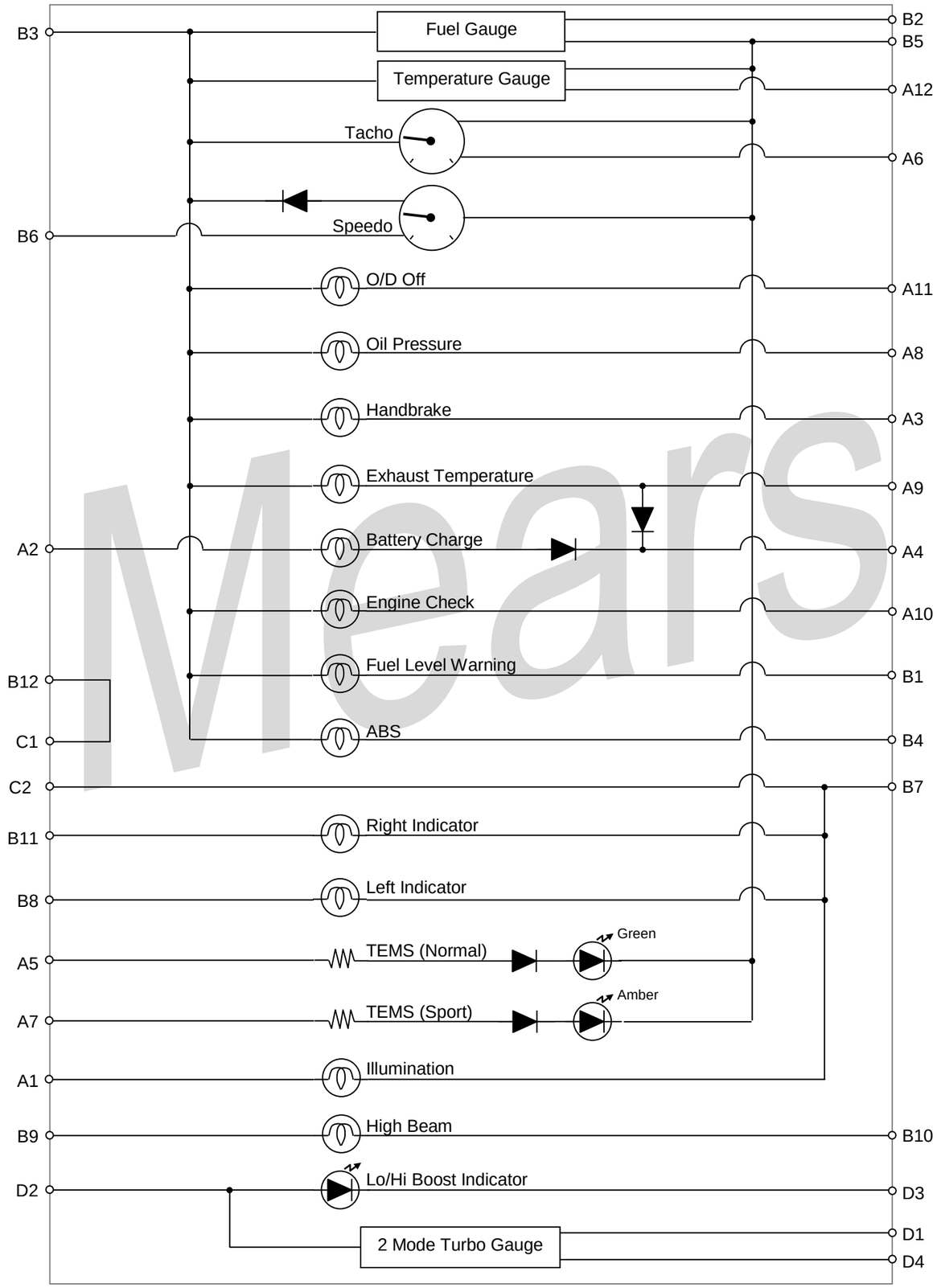


Fuel Level Warning Light (Ground)	B1	Mustard-Black (2 Silver dots on wire)
Fuel Level Gauge Signal	B2	Mustard-Blue
Ignition (From 'Dash' Fuse)	B3	Red-Blue
ABS (Ground)	B4	Red-Yellow (2 Silver dots on wire)
Ground (Gauges only)	B5	Brown
Speedo Signal	B6	Yellow-Blue (ECU Pin 'SPD')
Ground (Indicators & Lights)	B7	White-Black
LHS Indicator (Power)	B8	Green-Black
High Beam 12V (From 'H/Lights Right' Fuse)	B9	Red-Green
High Beam (Ground)	B10	Red-Yellow (1 Silver dot on wire) 'HU'
RHS Indicator (Power)	B11	Green-Yellow
A/T Reverse	B12	Red-Black



Boost Signal	D1	Yellow (Connects to C-18 Pin 3)
Ground	D2	Black (Connects to C-18 Pin 4)
2 Mode (Lo/Hi Boost Switch)	D3	Blue (Connects to C-18 Pin 2)
Ignition (IG+)	D4	Red (Connects to C-18 Pin 1)

Automatic Transmission Connector – C1 & C2



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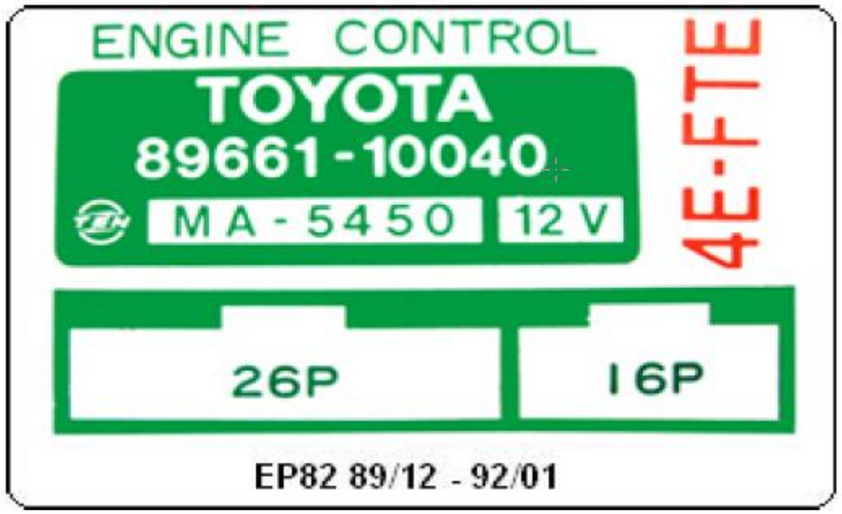
NOTE : The details on this page have been sourced from publicly accessible PDF detailing all EP82 & EP91 ECU's and pinouts http://street-legal-starlet.nl/Sagittarius/ecu's.pdf. The details are copyrighted to StreetLegalStarlet forums http://www.street-legal-starlet.nl/forum/, documented by Sagittarius®. These details are taken as a screenshot of the single PDF page relating to the Gen1 (Dec 1989 – Jan 1992) EP82 Toyota Starlet GT. ECU details for later models can be obtained from the PDF.



Green Sticker
P/N: 89661-10040
MA - 5450

EP82 AT + MT (both same ecu)
(89/12 - 92/01)

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All pinouts archived and checked with the original ecu's by Sagittarius®

E01	#10	STA	OX	G-	G1	IGF	IGT	THA	PIM	THW	NSW	
E02	#20	E1	KNK	E21	NE	G2	IDL	VC	PSW	E2	VSV1	DISC

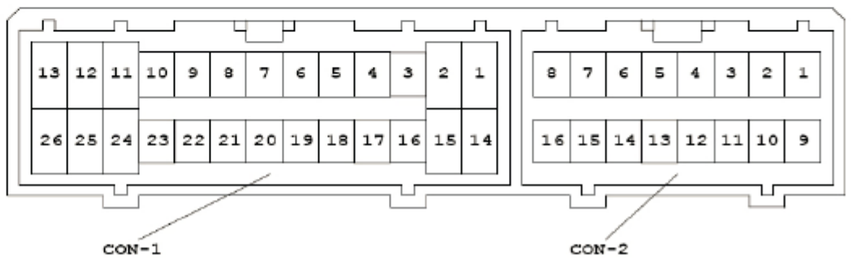
T	BC	AC2	ELS	FC	CCO	BATT	+B1
VF	AC1	ACT	SPD		EGW	W	+B

Connector 1

pin	name	function
1	---	---
2	NSW	Neutral Starter Switch
3	THW	Water temp signal
4	PIM	Pressure Intake Manifold (boost or vacuum)
5	THA	Intake Air temp
6	IGT	Ignition Timing
7	IGF	Ignition Feedback
8	G1	Distributor
9	G-	Distributor Crank Angle
10	OX	Oxygen sensor (lambda)
11	STA	Engine Cranking Signal (starter switch)
12	#10	Injector signal 1+3
13	E01	Computer Ground
14	DISC	Aux. Control Valve or Idle-Up System Vacuum Switching Valve
15	VSV1	Air Control Vacuum Switching Valve (hi-lo boost solenoid)
16	E2	Sensor Ground
17	PSW	Throttle Position Sensor (VTA = PSW)
18	VC	Throttle Position Sensor (pin 1) (also known as VCC)
19	IDL	Throttle Position Sensor (Signals ECU throttle position is at idle)
20	G2	Distributor
21	NE	Part of the IIA (Intergrated Ignition Assembly) (RPM)
22	E21	Sensor Ground
23	KNK	Knock Sensor
24	E1	Computer Ground
25	#20	Injector signal 2+4
26	E02	Computer Ground

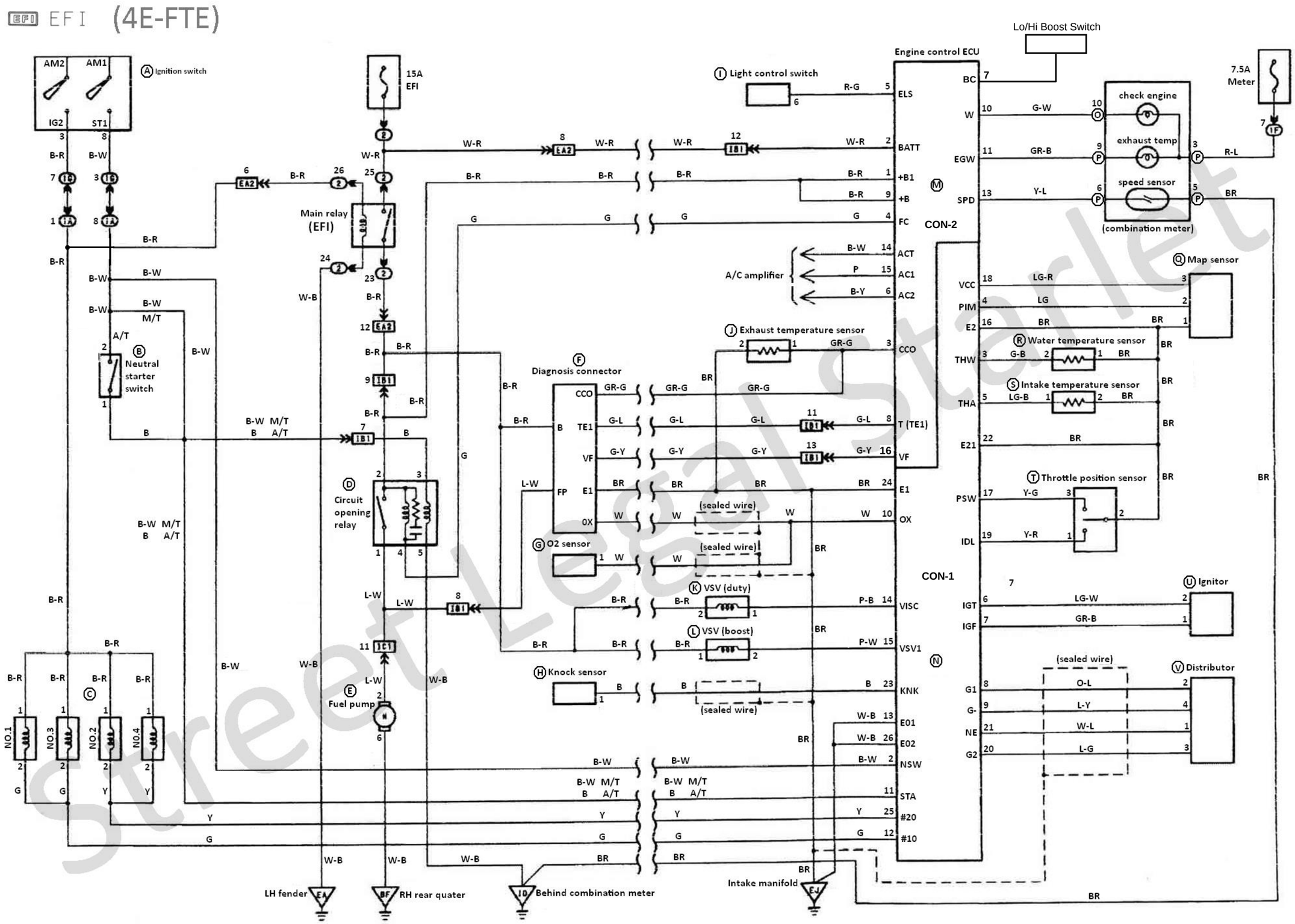
Connector 2

pin	name	function
1	+B1	Switched Battery Power (12V)
2	BATT	Switched Battery Power (+12V) (EFI Main Relay)
3	CCO	Check Connector Outputs
4	FC	Circuit Opening Relay (Fuel Cut)
5	ELS	Electrical Load Signal System Taillight Relay
6	AC2	12V To Aircon Computer
7	BC	Boost Control
8	T	Check Connector (Same as TE1)
9	+B	Switched Battery Power (12V)
10	W	Warning Light
11	EGW	Catalytic Converter Temparature Warning Light (Combination Meter)
12	---	---
13	SPD	Speed Signal
14	ACT	A/C Cut Control System
15	AC1	A/C Switch Signal System (12V to Aircon Computer)
16	VF	Check Connector Outputs

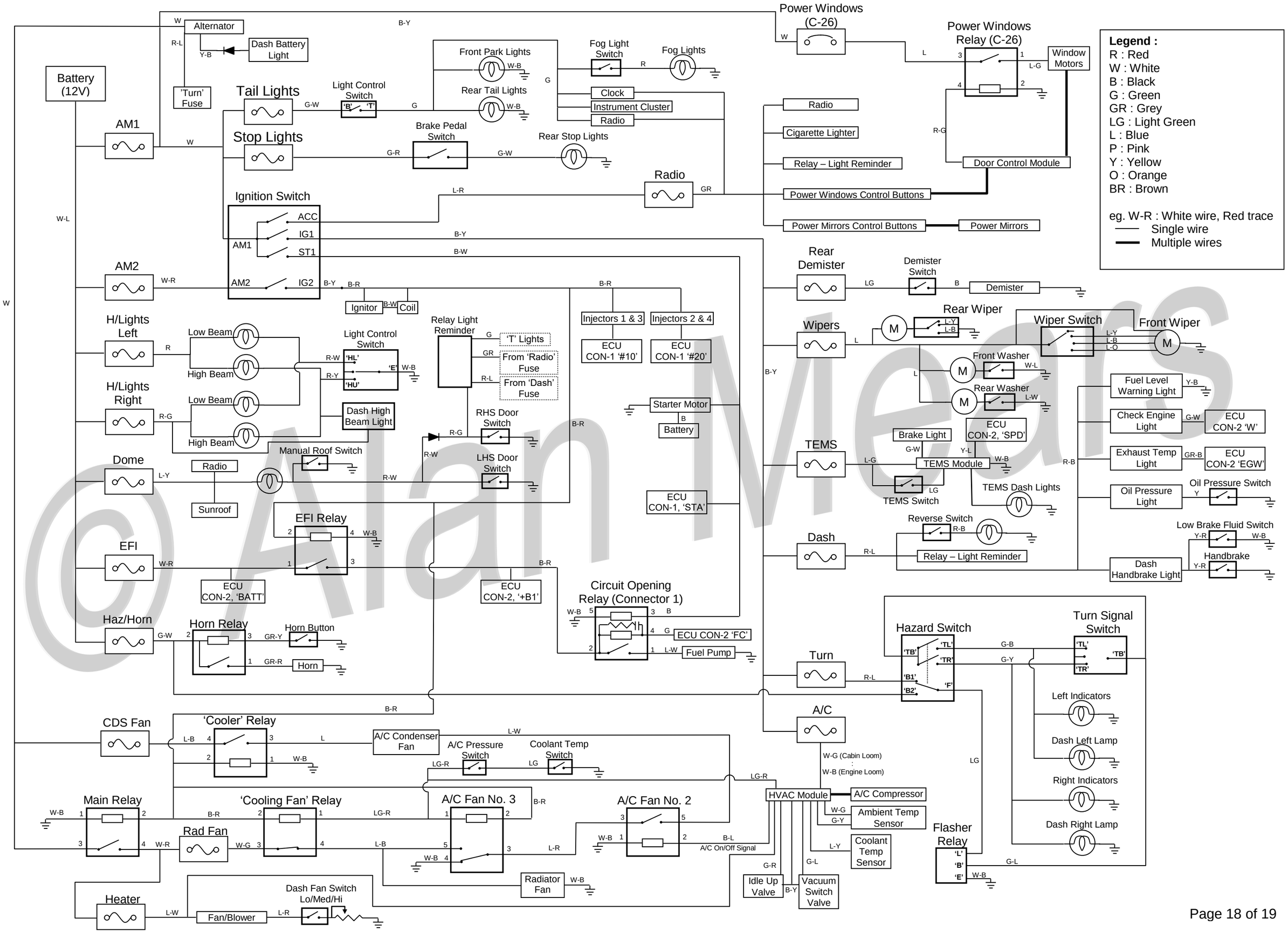


Street Legal Starlet

NOTE : The details on this page have been sourced from publicly accessible JPG detailing the EP82 (Zenki) ECU wiring diagram. The original JPG is copyrighted to StreetLegalStarlet forums http://street-legal-starlet.nl/Sagittarius/Wiring/SLS%20-%20EP82%20Wiring%20diagram%20%28zenki%29.jpg. Corrections made to the original JPG : 'VF' Pin number corrected from CON-2 Pin 15 to 16; CON-2 Pin 7 'BC' added; Coil input for Main EFI relay moved from 'ST1' to 'IG2' line 7



Street Legal Starlet


Legend :
R : Red
W : White
B : Black
G : Green
GR : Grey
LG : Light Green
L : Blue
P : Pink
Y : Yellow
O : Orange
BR : Brown
eg. W-R : White wire, Red trace
Single wire
Multiple wires
Battery (12V)
Alternator
Dash Battery Light
'Turn' Fuse
AM1
AM2
Tail Lights
Stop Lights
Light Control Switch
Front Park Lights
Rear Tail Lights
Fog Light Switch
Fog Lights
Clock
Instrument Cluster
Radio
Brake Pedal Switch
Rear Stop Lights
Ignition Switch
ACC
IG1
ST1
IG2
Ignitor
Coil
Power Windows (C-26)
Power Windows Relay (C-26)
Window Motors
Door Control Module
Radio
Cigarette Lighter
Relay – Light Reminder
Power Windows Control Buttons
Power Mirrors Control Buttons
Power Mirrors
Rear Demister
Demister Switch
Demister
Wipers
Rear Wiper
Wiper Switch
Front Wiper
Front Washer
Rear Washer
H/Lights Left
H/Lights Right
Low Beam
High Beam
Light Control Switch
Relay Light Reminder
'T' Lights
From 'Radio' Fuse
From 'Dash' Fuse
Dash High Beam Light
RHS Door Switch
LHS Door Switch
Manual Roof Switch
Dome
Sunroof
Injectors 1 & 3
Injectors 2 & 4
ECU CON-1 '#10'
ECU CON-1 '#20'
Starter Motor
Battery
ECU CON-1, 'STA'
TEMS
Brake Light
ECU CON-2, 'SPD'
TEMS Module
TEMS Switch
TEMS Dash Lights
Fuel Level Warning Light
Check Engine Light
ECU CON-2 'W'
Exhaust Temp Light
ECU CON-2 'EGW'
Oil Pressure Light
Oil Pressure Switch
Low Brake Fluid Switch
Handbrake
Dash Handbrake Light
Dash
Reverse Switch
Relay – Light Reminder
EFI
EFI Relay
ECU CON-2, 'BATT'
ECU CON-2, '+B1'
Circuit Opening Relay (Connector 1)
ECU CON-2 'FC'
Fuel Pump
Haz/Horn
Horn Relay
Horn Button
Horn
Turn
Hazard Switch
Turn Signal Switch
Left Indicators
Dash Left Lamp
Right Indicators
Dash Right Lamp
Flasher Relay
A/C
W-G (Cabin Loom)
W-B (Engine Loom)
HVAC Module
A/C Compressor
Ambient Temp Sensor
Coolant Temp Sensor
Idle Up Valve
Vacuum Switch Valve
A/C On/Off Signal
CDS Fan
'Cooler' Relay
A/C Condenser Fan
A/C Pressure Switch
Coolant Temp Switch
Main Relay
'Cooling Fan' Relay
Rad Fan
A/C Fan No. 3
A/C Fan No. 2
Radiator Fan
Heater
Fan/Blower
Dash Fan Switch Lo/Med/Hi

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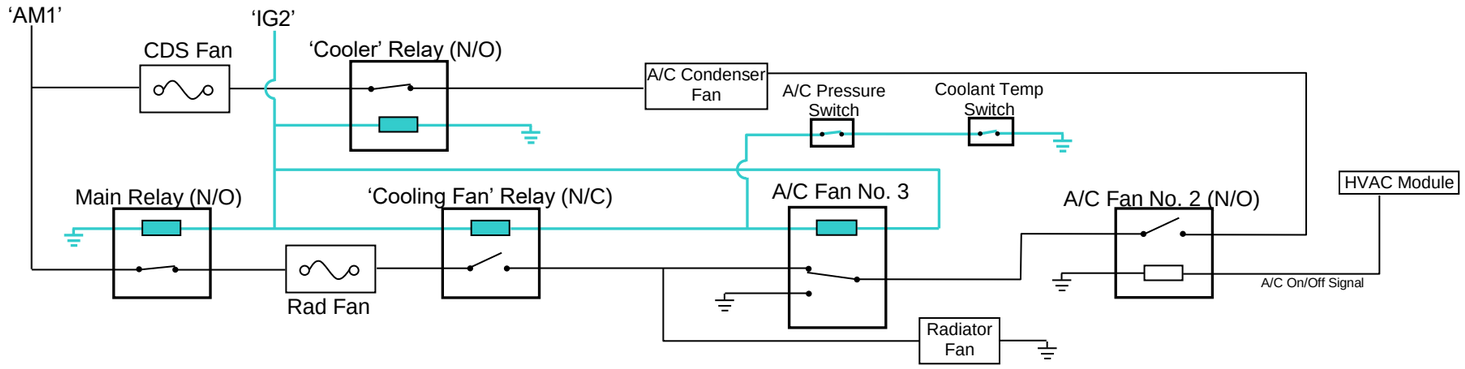
A/C Condenser Fan & Radiator Fan Relay Configurations

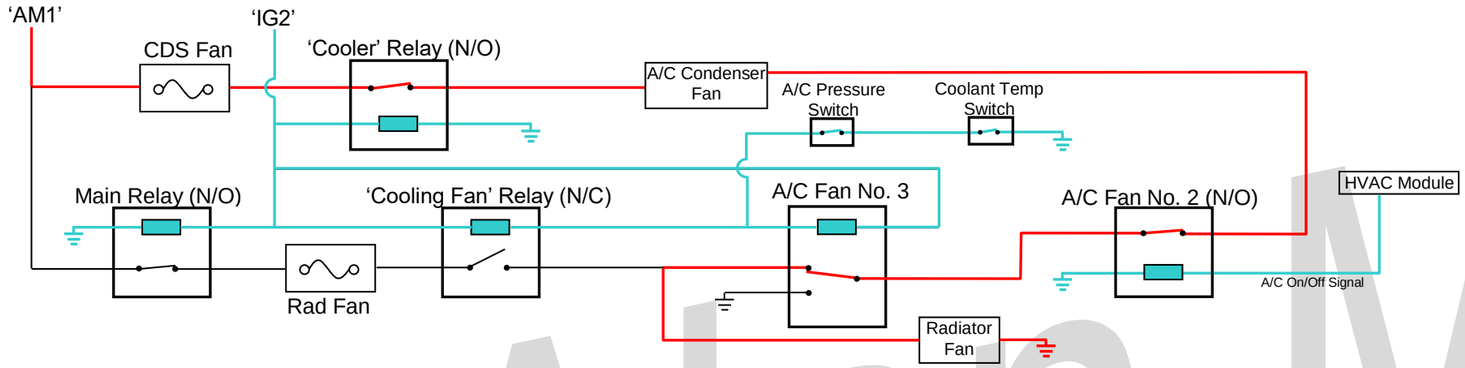
This page details the current flow through the relay coils and switches in various configurations

Legend :
- Coil Energised
- Current Flow through relay coils
- Current Flow through relay contacts



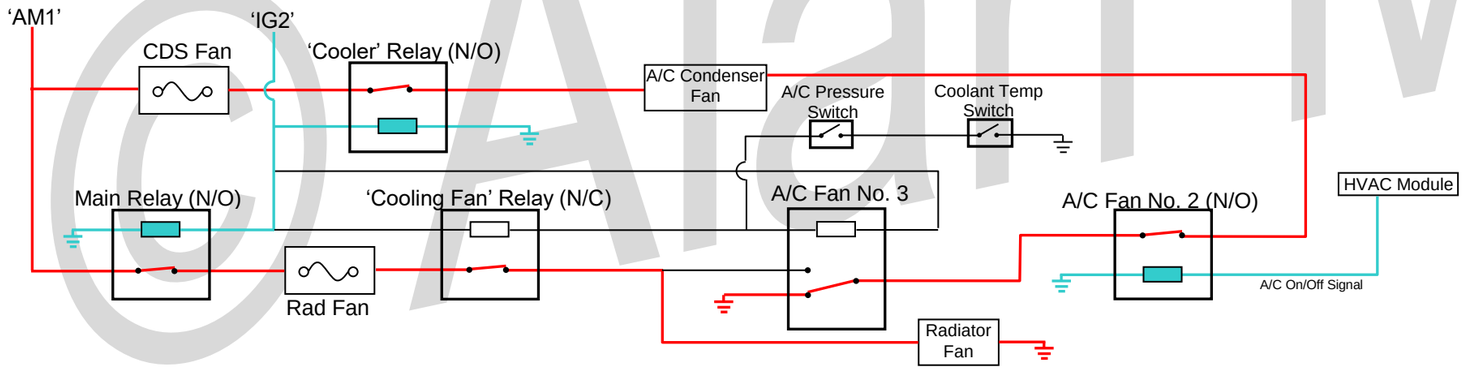
- Ignition On
- A/C Off
- Normal Coolant Temperature
- Normal A/C Pressure

No fans running



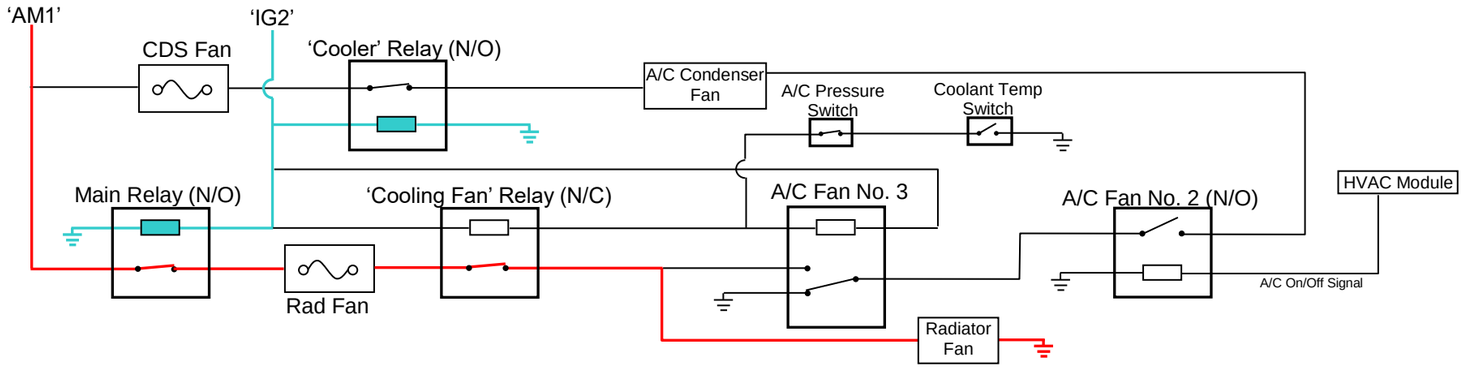
- Ignition On
- A/C On
- Normal Coolant Temperature
- Normal A/C Pressure

Both fans powered in series and both run at low speed



- Ignition On
- A/C On
- High Coolant Temperature or High A/C Pressure

Fans powered in parallel through separate fuses and run at high (full) speed



- Ignition On
- A/C Off
- High Coolant Temperature
- Normal A/C Pressure

A/C Condenser fan not running. Radiator Fan runs as high (full) speed

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